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Report No: ICR00057

IMPLEMENTATION COMPLETION AND RESULTS REPORT
(Credit No. IDA-58370)

ON A

CREDIT

IN THE AMOUNT OF SDR 55.2 MILLION
(US$75.2 MILLION EQUIVALENT)

TO THE
PEOPLE's REPUBLIC OF BANGLADESH

FOR
Bangladesh Weather and Climate Services Regional Project (150220)
May 28, 2025

Urban, Resilience, and Land
South Asia

CURRENCY EQUIVALENTS

(Exchange Rate Effective May 12, 2025)

Currency Unit =	Bangladesh Taka (BDT)
US$1.00 =	BDT 122
BDT 1.00 =	US$ 0.008
US$1.00 =	SDR 0.745

FISCAL YEAR
July 1 - June 30

Regional Vice President:	Martin Raiser
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ABBREVIATIONS AND ACRONYMS

AgAWS	Agricultural Automatic Weather Station
AIS	Agricultural Information Service
ARG	Automatic Rain Gauge
AWOS	Aviation Weather Observation System
AWS	Automatic Weather Stations
BAMIS	Bangladesh Agrometeorological Information System
BARI	Bangladesh Agricultural Research Institute
BCR	Benefit-Cost Ratio
BJRI	Bangladesh Jute Research Institute
BMD	Bangladesh Meteorological Department
BRRI	Bangladesh Rice Research Institute
BSRI	Bangladesh Sugarcrop Research Institute
BWCSRP	Bangladesh Weather and Climate Services Regional Project
BWDB	Bangladesh Water Development Board
CBA	Cost-Benefit Analysis
CPP	Cyclone Preparedness Programme
DA	Designated account
DAE	Department of Agricultural Extension
DDM	Department of Disaster Management
DDS	DCP Data Service
DEM	Digital elevation model
DPD	Deputy Project Director
EMF	Environmental Management Framework
EMP	Environmental Management Plan
EWS	Early Warning System
FAPAD	Foreign Aided Project Directorate
FFGS	Flash Flood Guidance System
FFWC	Flood Forecasting and Warning Center
FM	Financial Management
GAAP	Governance and Accountability Action Plan
GEO	Group on Earth Observations
GIS	Geographic Information System
GoB	Government of Bangladesh
GPS	Global Positioning System
GTS	Global Telecommunication System (WMO)
HPC	High-Performance Computing
ICT	Information and Communication Technologies
IDA	International Development Association
IMD	India Meteorological Department
IE	Implementation Entity
IT	Information Technology
ICR	Implementation Completion Report
ICT	Information and Communication Technology
ISR	Implementation Status Report
M&E	Monitoring and evaluation
O&M	Operations and Maintenance
PD	Project Director
PDO	Project Development Objective
PIU	Project Implementation Unit
PV	Present Value
PWS	Public Weather Services
QPE	Quantitative Precipitation Estimation
QPF	Quantitative Precipitation Forecast
SAHF	South Asia Hydromet Forum
SAR	South Asia Region
SASCOF	South Asian Climate Outlook Forum
SIM	Subscriber Identity Module
TOR	Terms of Reference
WMO	World Meteorological Organization
WTP	Willingness to Pay

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DATA SHEET

BASIC DATA

Product Information

Operation ID P150220	Operation Name Bangladesh Weather and Climate Services Regional Project
Product Investment Project Financing (IPF)	Operation Short Name BWCSRP
Operation Status Closed	Approval Fiscal Year 2016
Original EA Category Partial Assessment (B) (Approval package - 03 Jun 2016)	Current EA Category Partial Assessment (B) (Restructuring Data Sheet - 30 Nov 2024)

CLIENTS

Borrower/Recipient People's Republic of Bangladesh	Implementing Agency Bangladesh Meteorology Department (BMD), Bangladesh Water Development Board (BWDB), Department of Agricultural Extension (DAE)

DEVELOPMENT OBJECTIVE

Original Development Objective (Approved as part of Approval Package on 03-Jun-2016)

The Project Development Objective of this project is “to strengthen Government of Bangladesh’s capacity to deliver reliable weather, water and climate information services and improve access to such services by priority sectors and communities.”

FINANCING

Financing Source	Original Amount (US$)	Revised Amount (US$)	Actual Disbursed (US$)
World Bank Financing	113,000,000.00	73,779,274.20	75,285,779.00
IDA-58370	113,000,000.00	73,779,274.20	75,285,779.00
Non-World Bank Financing	14,800,000.00	8,239,998.00	8,239,998.00
Borrower/Recipient	14,800,000.00	8,239,998.00	8,239,998.00
Total	**127,800,000.00**	**82,019,272.20**	**83,525,777.00**

RESTRUCTURING AND/OR ADDITIONAL FINANCING

Date(s)	Type	Amount Disbursed (US$M)	Key Revisions
31-Jan-2019	Portal	9.57	• Reallocations
17-May-2020	Portal	22.56	• Components • Results • Risks • Disbursement Estimates • Loan Cancellations • Reallocations • Other Changes
14-Jun-2022	Portal	43.81	• Loan Closing Date Extension
08-May-2023	Portal	58.79	• Components • Loan Cancellations • Reallocations
19-Dec-2023	Portal	65.72	• Loan Closing Date Extension
30-Nov-2024	Portal	74.20	• Results • Disbursement Estimates • Loan Cancellations • Reallocations

KEY DATES

Key Events	Planned Date	Actual Date
Concept Review	04-Feb-2015	08-Feb-2015
Decision Review	01-Dec-2015	01-Dec-2015

Authorize Negotiations	02-May-2016	03-May-2016
Approval	03-Jun-2016	03-Jun-2016
Signing	05-Apr-2017	05-Apr-2017
Effectiveness	04-May-2017	04-May-2017
Restructuring Sequence.01	Not Applicable	31-Jan-2019
Restructuring Sequence.02	Not Applicable	17-May-2020
Restructuring Sequence.03	Not Applicable	14-Jun-2022
Restructuring Sequence.04	Not Applicable	08-May-2023
Restructuring Sequence.05	Not Applicable	19-Dec-2023
Restructuring Sequence.06	Not Applicable	30-Nov-2024
Mid-Term Review No. 01	26-Jan-2020	30-Jan-2020
Operation Closing/Cancellation	30-Nov-2024	30-Nov-2024
ICR/NCO	30-May-2025	--

RATINGS SUMMARY

Outcome	Bank Performance	M&E Quality
Satisfactory	Satisfactory	Substantial

ISR RATINGS

No.	Date ISR Archived	DO Rating	IP Rating	Actual Disbursements (US$M)
01	21-Oct-2016	Satisfactory	Satisfactory	0.00
02	23-May-2017	Satisfactory	Satisfactory	0.00
03	01-Dec-2017	Satisfactory	Satisfactory	4.67
04	15-Jun-2018	Satisfactory	Satisfactory	7.15
05	20-Dec-2018	Satisfactory	Moderately Satisfactory	9.42
06	12-Jul-2019	Moderately Unsatisfactory	Moderately Unsatisfactory	16.53

07	11-Mar-2020	Moderately Unsatisfactory	Moderately Unsatisfactory	22.10
08	30-Jun-2020	Moderately Satisfactory	Moderately Satisfactory	24.75
09	06-Dec-2020	Moderately Satisfactory	Moderately Satisfactory	27.41
10	06-Oct-2021	Moderately Satisfactory	Moderately Satisfactory	34.90
11	03-Apr-2022	Moderately Satisfactory	Moderately Satisfactory	37.54
12	08-Oct-2022	Moderately Satisfactory	Moderately Satisfactory	52.36
13	30-Mar-2023	Satisfactory	Satisfactory	57.04
14	29-Nov-2023	Moderately Satisfactory	Moderately Satisfactory	65.09
15	16-Jun-2024	Satisfactory	Satisfactory	70.56
16	30-Nov-2024	Satisfactory	Satisfactory	74.20

SECTORS AND THEMES

Sectors

Major Sector	Sector	%	Adaptation Co-benefits (%)	Mitigation Co-benefits (%)
FY17 - Agriculture, Fishing and Forestry	FY17 - Other Agriculture, Fishing and Forestry	8	100	0
	FY17 - Public Administration - Agriculture, Fishing & Forestry	4	100	0
FY17 - Water, Sanitation and Waste Management	FY17 - Other Water Supply, Sanitation and Waste Management	68	100	0
	FY17 - Public Administration - Water, Sanitation and Waste Management	20	100	0

Themes

Major Theme	Theme (Level 2)	Theme (Level 3)	%
FY17 - Environment and Natural Resource Management	FY17 - Water Resource Management	FY17 - Water Institutions, Policies and Reform	18
FY17 - Finance	FY17 - Finance for Development	FY17 - Disaster Risk Finance	21

FY17 - Urban and Rural Development	FY17 - Disaster Risk Management	FY17 - Disaster Preparedness	21
		FY17 - Disaster Response and Recovery	21
		FY17 - Disaster Risk Reduction	21

ADM STAFF

Role	At Approval	At ICR
Practice Manager	Bernice K. Van Bronkhorst	Abhas Kumar Jha
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ICR Main Contributor	Tjark Gall	

I. PROJECT CONTEXT AND DEVELOPMENT OBJECTIVES

A. CONTEXT AT APPRAISAL

1. **Country Context:** At appraisal, Bangladesh ranked among the most climate-vulnerable countries, facing increasing climate variability and extreme weather events such as floods and cyclones. Bangladesh's low-lying deltaic geography and high population density make it highly vulnerable to weather-related hazards, causing significant economic losses, particularly in agriculture and fisheries. Between 1980 and 2010, climate-related disasters affected over 200 million people and caused cumulative damages exceeding US$12 billion. The 2010 Bangladesh Climate Change Strategy and Action Plan (BCCSAP) highlighted the need for enhanced hydrometeorological (hydromet) services, forecasting capacity, and early warning systems to support climate resilience and disaster preparedness.
2. **Hydro- and Agromet Sector Contexts:** Bangladesh's hydromet services were fragmented, outdated, and underfunded, limiting their effectiveness in disaster risk management and climate-sensitive economic sectors. The Bangladesh Meteorological Department (BMD) and the Bangladesh Water Development Board (BWDB) operated with limited real-time observation networks, outdated forecasting models, and inadequate data-sharing mechanisms. Low forecasting accuracy and insufficient lead times for extreme weather warnings hindered informed decision-making by farmers and emergency responders. Agromet did not exist as a formalized service, and the agricultural sector, which employed nearly 40 percent of the population, lacked localized climate services, making it highly vulnerable to unpredictable weather patterns. Flood and coastal inundation early warning systems were limited in coverage and accessibility, leaving millions without reliable information to prepare for monsoon flooding and storm surges. An integrated approach to hydro- and agromet services is essential for effective risk management.
3. **Rationale for World Bank Engagement:** The World Bank engaged in the BWCSRP to modernize Bangladesh's hydromet sector and enhance climate resilience.[1] The Bank's experience in strengthening early warning systems (EWS), climate services, and disaster risk management in South Asia positioned it as a key partner in addressing the country's institutional and technical gaps. The BWCSRP sought to provide a comprehensive modernization of weather, water, and climate services, focusing on improving forecasting accuracy, expanding early warning production and dissemination, and strengthening institutional capacity. The Project also aimed to enhance service delivery for climate-sensitive sectors, particularly agriculture, by developing the Bangladesh Agrometeorological Information System (BAMIS) and aviation through improved local weather monitoring. By integrating hydro- and agromet forecasting, the Project was designed to reduce climate-related losses, improve decision-making, and increase resilience at institutional and community levels.
4. **At appraisal, the Project was 'fully consistent with the World Bank Group's Country Partnership Framework [CPF] for Bangladesh for 2016–2020[2].** By supporting improved weather, water, and climate information services and disaster-related EWS, the Project contributed to the CPF's priority focus on strengthening disaster and climate resilience, and supported the broader efforts to promote regional cross-border cooperation on resilience.' (p. 6–7).

A.1. THEORY OF CHANGE (RESULTS CHAIN)

[1] Initial project financing included US$82 million equivalent national IDA and US$31 equivalent regional IDA due to co-benefits for regional data sharing, knowledge exchange, and improvements of weather observation equipment at three international airports.
[2] Bangladesh CPF for FY16–20, discussed by the Board of Executive Directors on April 5, 2016 (Report No. 103723-BD).

5. **A Theory of Change (ToC) was created ex-post** (Figure 1). It builds on a formalized problem statement[3], as well as objectives, outcomes, components, and activities described in the project documents. Restructurings only led to limited changes, e.g., reducing the number of stations or scope of technical consultancies, and a few activities being dropped (Doppler Weather Radars and Coastal-Marine Automated Network), which are highlighted in the ToC. Therefore, the ToC reflects the Project at both appraisal and closing.

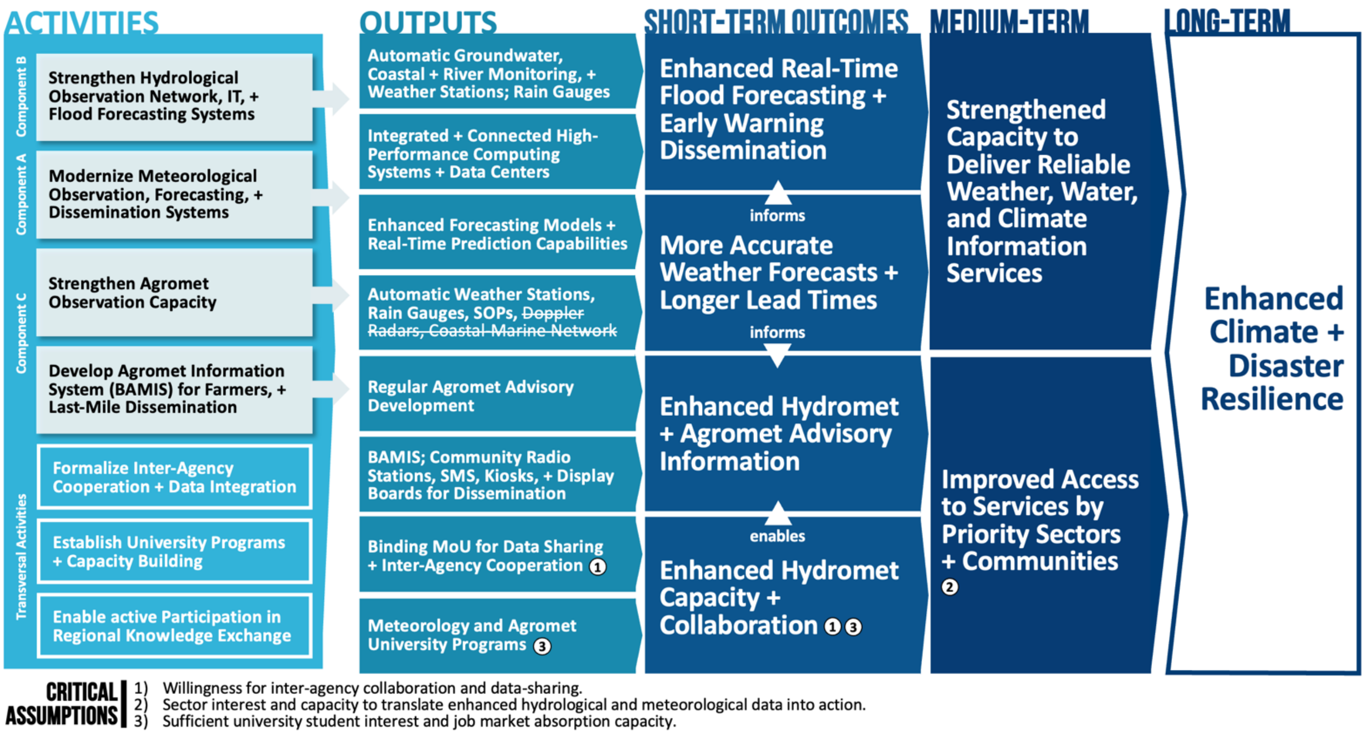

Figure 1: Project's Ex Post Theory of Change

A.2. PROJECT DEVELOPMENT OBJECTIVE (PDO)

6. The PDO at appraisal was **'to strengthen Bangladesh's capacity to deliver reliable weather, water and climate information services and improve access to such services by priority sectors and communities'** and has not been changed during the project implementation period.

A.3. KEY EXPECTED OUTCOMES AND OUTCOME INDICATORS

7. **The PDO is structured around two outcomes, which are reflected in the three PDO-level results indicators** (Table 1). The first outcome focuses on strengthening Bangladesh's capacity to deliver reliable weather, water, and climate information services, which is measured through improvements in forecasting capabilities (targeting BMD and BWDB). The second outcome emphasizes improving access to such services for *priority sectors and communities*, ensuring that key stakeholders receive timely and actionable climate information. The *sectors* refer to (i) water, (ii) disaster risk management, (iii) aviation, and (iv) agriculture and *communities* to professional users, such as forecasters and end-users/beneficiaries, i.e., farmers. While the formulation of the second outcome is broad with

[3] Bangladesh's intensifying climate change impacts combined with lack of adequate observation networks and forecasting capacity results in substantial risk on communities and priority sectors such as agriculture and water management.

'priority sectors and communities', the second and third PDO-level results indicators cover this large scope, and overall, the indicators are sufficient to measure key aspects of the project development outcomes. This is further supported by the fact that the PDO-level indicators remained essentially unchanged (only PDO1 was revised to ensure its measurability).

TABLE 1: PROJECT OUTCOMES AND PDO-LEVEL RESULTS INDICATORS

Expected Outcomes	PDO-Level Results Indicators
O1: Strengthened Bangladesh's capacity to deliver reliable *weather, water, and climate* information services	PDO1: Improved capacity for forecasting
O2: Improved access to such services by *priority sectors and communities*	PDO2: Improved service delivery to targeted agriculture sector beneficiaries.
	PDO3: Systematic measurement of user satisfaction in place to measure improvements in weather information service delivery: Establishment of a user satisfaction measurement system (Yes/No)

A.4. COMPONENTS

8. **Component A: Strengthening Meteorological Information Services (original cost: US$67.5 million, actual cost: US$24.3; Implementing Entity [IE]: BMD; partner organization: University of Dhaka, Department of Meteorology).** This component aimed to modernize the meteorological observation and forecasting capabilities of Bangladesh, enhancing the accuracy, reliability, and accessibility of weather and climate services. It aimed to strengthen BMD's meteorological infrastructure by upgrading surface, marine, and upper-air observation networks and enhancing numerical weather prediction systems. The Project aimed to establish automated meteorological stations, expand aviation weather observation capabilities, and integrate satellite data for improved monitoring and forecasting accuracy. Additionally, it sought to improve BMD's data-sharing mechanisms with regional and global weather networks to enhance early warning capacities. A key focus was strengthening institutional capacity through targeted training and improving decision-support systems and communication platforms for better weather forecast dissemination. The component also aimed to promote regional collaboration with neighboring countries to enhance disaster preparedness through data exchange and coordinated forecasting efforts and included partnerships with the University of Dhaka to upgrade their facilities and establish study programs in the meteorology department.

9. **Component B: Strengthening Hydrological Information Services and Early Warning Systems (original cost: US$45.30 million, actual cost: US$35.6; IE: BWDB).** This component was designed to modernize Bangladesh's hydrological observation and flood forecasting systems to improve water resource management and disaster preparedness. It aimed to expand and upgrade BWDB's hydrological observation network by installing automated water level gauges, rainfall monitoring stations, and real-time data transmission systems. The Project sought to develop more advanced hydrological modeling tools to predict flood risks with greater accuracy, particularly in urban areas and along major river basins. A key objective was to enhance early warning capabilities by integrating hydrological data into a national disaster response system for timely and localized flood alerts. To ensure sustainability, the component included technical training programs for BWDB staff, measures to improve institutional coordination among government agencies, and enhanced public awareness campaigns on flood

preparedness. Additionally, the Project planned to strengthen transboundary water data-sharing with neighboring countries to improve regional flood forecasting and expand weather observation at international airports with regional and global benefits for security and accessibility.

10. **Component C: Agrometeorological Information Systems Development (original cost: US$16.1 million, actual cost: US$21.1; IE: Department of Agricultural Extension [DAE]; four research agencies – Bangladesh Agricultural Research Institution [BARI], Bangladesh Rice Research Institution [BRRI], Bangladesh Sugarcrop Research Institute [BSRI], and Bangladesh Jute Research Institution [BJRI]; two Agricultural Universities – Bangladesh Agricultural University [BAU] and Bangabandhu Sheikh Mujibur Rahman Agricultural University [BSMRAU]).** This component sought to enhance weather and climate information services for farmers, thereby improving agricultural resilience and productivity. It was designed to develop the Bangladesh Agrometeorological Information System (BAMIS), a national platform for integrating meteorological and agricultural data to generate tailored advisories for different agro-ecological zones. The Project sought to install a network of agricultural automatic weather stations (AWS) to collect data on temperature, humidity, soil moisture, and other critical variables. It also aimed to develop forecasting models for agricultural droughts linked with meteorological droughts, floods, and seasonal climate patterns to help farmers make more informed decisions on planting, irrigation, and harvesting. A key element of this component was the training of agricultural extension officers in interpreting and disseminating agrometeorological advisories to farmers through digital platforms, SMS services, and community outreach programs. The Project planned to establish feedback mechanisms, such as farmer surveys, to refine advisory services continuously. Close collaboration with research institutions was part of Component C, including the establishment of master's programs in agro-meteorology at BAU and BSMRAU.

11. **Component D: Contingent Emergency Response Component (CERC) (original cost: US$0.0, actual cost: US$0.0; IE: n.A.).** This component was included as a flexible mechanism to provide rapid financing for emergency response and recovery efforts in the event of a disaster but was not activated.

B. SIGNIFICANT CHANGES DURING IMPLEMENTATION

12. **The Project underwent six restructurings between 2019 and 2024, but none did alter the PDO or components significantly**. The following restructurings and cancelations were conducted:

(a) *First Restructuring (February 2, 2019):* This restructuring corrected inconsistencies between the Financing Agreement and the Project Appraisal Document (PAD) by adding civil works as eligible expenses.

(b) *Second Restructuring (April 29, 2020):* Following initial delays and low disbursements during the first two years, the project scope was reduced by canceling SDR 17.67 million in the IDA credit (equivalent to US$24 million) and reallocating funds across components. Components A and B were scaled down, especially regarding the number of monitoring stations and canceling elements that were no longer needed (see next section for details). At the same time, Component C was expanded due to its strong performance. The restructuring also revised intermediate results indicators and adjusted implementation priorities.

(c) *Third Restructuring (June 7, 2022):* The Project was extended by 12 months to December 31, 2023, to allow for the completion of critical activities that were delayed due to an extended period of nearly one year between Board approval and effectiveness, COVID-19, technical challenges, and procurement bottlenecks.

(d) *Fourth Restructuring (April 26, 2023):* A partial cancellation of SDR 2.34 million (equivalent to US$2.85 million) was processed. The cancelation was driven by savings from Component A, which resulted from changes in procurement strategies and adjustments in BMD's operational priorities.

(e) *Fifth Restructuring (December 18, 2023):* The Project was extended by an additional 11 months, shifting the closing date to November 30, 2024. This extension was granted to allow for the final integration, testing, and troubleshooting of hydromet systems during the 2024 monsoon season. It ensured that all systems would be operational and fully validated before the Project's conclusion.

(f) *Sixth Restructuring (November 2024):* The remaining funds of the Project were canceled (SDR 4.54 million, equivalent to US$6.05 million), and the closing dates in the results framework were corrected.

B.1. REVISED COMPONENTS

13. **There were no significant changes to the component setup; however, a scope reduction occurred as part of the second and fourth restructurings.**

(a) <u>Component A:</u> The second restructuring involved reducing the number of observation stations and high-performance computing capacity. It also led to dropping related technical studies like storm surge modeling to match BMD's implementation capacity. Several technical consultancies were streamlined or integrated into broader contracts. Consequently, packages such as operational research, satellite imagery, and data-sharing software were dropped due to shifting agency priorities.

(b) <u>Component B:</u> The second restructuring adjusted the number of observation stations across all types (e.g., 344 to 308 surface water level stations, 272 to 257 rain gauges, 1,100 to 905 ground water stations, 10 to 3 climate stations) and modified the Information and Communication Technology (ICT) setup. Some technical studies and early warning system investments were scaled down or dropped.

(c) <u>Component C:</u> The second restructuring expanded the scope to include support to agromet departments at two universities, upgrades to the agromet portal and decision-support systems, and new measurement stations. Dissemination was enhanced through community radio, digital displays, and additional advisory content.

B.2. REVISED PDO, OUTCOME TARGETS, AND RESULTS INDICATORS

14. **The PDO itself remained unchanged throughout the implementation period.** The second restructuring led to adjustments in results indicators and target values to align with the changed project scope. At that time, US$22.47 million out of a total of US$75.21 million (30 percent) had been disbursed. No split rating was performed since most indicators, and some target values were only adapted slightly or made more explicit. Further, multiple indicators are text-based, limiting the ability for direct comparisons.

(a) *PDO-level results indicator 1* targets were changed from '25% Increase of skill over baseline' to 'Improvement in forecasts by at least one attribute (skill/lead time/frequency/spatial range)' to better reflect the multidimensional improvement of forecasts. A detailed description of indicators was added.

(b) *PDO-level results indicator 2* targets were adjusted based on the actual lead farmers database – raising the target from 70 to 80, while lowering the target for female beneficiaries from 40 to 25 percent.

(c) *PDO-level results indicator 3* targets have been revised from 'User satisfaction survey is conducted' to 'Two Surveys are conducted (baseline and endline)' to add an additional survey.

(d) *Intermediate results indicators* were updated to reflect the reduced scope:

(i) IRI 1 ‘Upgraded and modernized hydromet related network established’ target was reduced from 70 to 60 percent.

(ii) IRI 2 was changed from ‘Strengthened capacity for delivery and use of weather and climate services’ with a target of ‘75% of targeted training completed’ to ‘Strengthened pipeline of meteorologists in country’ with a target of ’40 students (gender disaggregated).’

(iii) IRI 3 ‘Improved financial sustainability to maintain hydromet networks’ target was changed from 50 percent to ‘Post project operational plan approved, and implementation initiated.’

(iv) IRI 4 was changed from ‘Greater regional cooperation on weather services’ with target ‘Joint partnerships for regional product’ to ‘Improved regional dialogue and exchange on weather, water and climate services’ with target ‘Implementing agency participants engage in at least 3 regional dialogue meetings; 10 staff participate in regional trainings.’

(v) IRI 5 ‘Increase in weather, water and climate products developed’ target was changed from ‘Customized Agro advisories; Improved PWS; QPF; QPE; Nowcasting’ to ‘BMD: portal for improved Public weather services dissemination established; BWDB: water resource advisories and hydrological outlooks; DAE: customized agro-advisories.’

(vi) IRI 6 was changed from ‘Improved access to agromet information at the Upazila level’ with target ‘BAMIS is in place; Kiosks installed in 50% Upazilas’ to ‘Improved access to agromet information at the District and Upazila level’ with target ‘BAMIS is in place; District Display and Upazila Kiosks in 70% locations.’

(vii) IRI 7 ‘Community Early Warning Systems strengthened in 4 districts’ was dropped.

B.3. RATIONALE FOR CHANGES AND THEIR IMPLICATION ON THE ORIGINAL TOC

15. **Most changes were minor or related to project management, such as adjusting inconsistencies, extending deadlines, or canceling unallocated funds.** Only the second restructuring had a larger impact by rightsizing the Project after initial delays to prioritize key components and reduce the Project’s complexity. This included reducing the overall project funding envelope as well as reallocating funds to Component C due to good performance. The changes were justified by initial delays in procurement, institutional challenges such as hiring qualified staff, and a strategic shift toward more achievable objectives within the remaining project period. The PDO remained unchanged, and a few complex activities were removed, such as radars, leaving systemic outcomes and targets largely unaffected. While the forecasting accuracy could have been higher with more equipment, successful establishment and integration of a smaller but functional system was prioritized and achieved, and government support reaffirmed.

II. OUTCOME

A. RELEVANCE OF PDO

A.1. ASSESSMENT OF RELEVANCE OF PDO AND RATING

16. **The PDO remains critical for Bangladesh and the World Bank, and its relevance is consequently rated as 'High.'** The Project is consistent with the current CPF[4] high-level objective C 'enhanced climate resilience', 'informed by the SCD frontier challenge on addressing climate and environmental vulnerabilities while being aligned with the 8th FYP's theme of a sustainable development pathway resilient to disasters and climate change. The relevance of the agricultural sector is highlighted as well. 'Climate change poses a serious threat to the country's agricultural productivity growth, with projected rising heat, loss of agriculture land and increased flooding.'

17. **The current CPF further emphasizes the continuous priority of the World Bank to support Bangladesh's capacity to better manage climate disasters.** The current CPF mentions that the country has established policies and legal frameworks, strengthened institutions and systems, and enhanced hydrometeorological and forecasting capabilities, including community-based early warning systems – referring to BWCSRP. The CPF confirms that the World Bank 'will help the country further strengthen disaster risk management, including developing climate-resilient infrastructure, green affordable housing, and risk insurance mechanisms, including community-led adaptation,' for all of which reliable and accurate hydromet information is fundamental.

18. **As the first hydromet project in Bangladesh, the Project had a realistic level of ambition.** The similarity between the Project at design and completion shows that it had high but realistic ambitions and could address most of them with minor changes, e.g., the number of monitoring stations, resulting from changing priorities, sectoral challenges, or external factors, thus remaining relevant by adapting to the evolving needs and priorities. The Project has set foundations for further improvements and expansions of sectors supported with climate services, built the data and ICT infrastructure at key institutions (BMD and BWDB) to continuously improve their observational networks and forecasting capacity, and built capacity within organizations to maintain and expand the current quality and thus remains highly relevant.

B. ACHIEVEMENT OF PDOS (EFFICACY)

B.1. ASSESSMENT OF ACHIEVEMENT OF THE FIRST OBJECTIVE OF STRENGTHENING BANGLADESH'S CAPACITY TO DELIVER RELIABLE WEATHER, WATER, AND CLIMATE INFORMATION SERVICES

19. **The outcome of strengthening the forecasting capacity was fully achieved and is rated 'High.'** Improvements were noted in all forecasting attributes, including skill, lead time, frequency, and spatial resolution, confirmed by technical validation notes from BMD and BWDB. BMD increased its forecast resolution from 18 to 3 kilometers, extended lead times from one to three days, and expanded model runs from one to four times daily. BWDB scaled up its flood forecasting coverage from 31 to all 64 districts and developed additional models addressing groundwater, drought, salinity, and river morphology.

20. **Investments in technical modernization and institutional coordination directly improved forecasting performance, validating the project's approach during the preparation stage.** Project design connected investments in observational systems, ICT infrastructure, forecasting models, and inter-agency integration with measurable service delivery gains. These inputs were intended to create a shift from manual, fragmented systems toward an automated, real-time, and user-responsive hydromet service. The PAD also emphasized the importance of

[4] World Bank *Bangladesh – Country Partnership Framework for the Period FY2327,* Report No. 181003, discussed by the World Bank Board of Executive Directors on April 27, 2023.

institutional ownership and sustainability, with targeted support for standard operating procedures, training, and inter-agency data protocols.

21. **The Project provided extensive technical and financial input across all implementing agencies.** More than 1,900 observation stations were installed, including 224 by BMD (comprising synoptic AWS, Ag-AWS with soil moisture sensors, and ARGs), 905 groundwater monitoring stations, and 315 river water level stations by BWDB, and 159 Ag-AWS by DAE. Specialized procurement packages also equipped agencies with pilot and radiosondes, aviation weather observing systems (AWOS), calibration instruments, hydrogen generators, and sonar-based hydrological survey tools. BMD installed a high-performance computing (HPC) system, while BWDB and DAE upgraded data centers and field equipment. The Project financed training for staff, regional knowledge exchanges, and maintenance plans.

22. **These inputs strengthened forecasting systems, institutional capacity, and operational integration.** BMD operationalized a fully integrated meteorological information and communication system in a dedicated temperature and humidity-controlled room with fire safety measures, data backup, configuration for automatic recovery as well as 24/7 monitoring in a dedicated Network Operations Center. Data from all observation networks flow into the hardware/software system, are stored, transmitted to the HPC for the running of different models, and outputs are passed on to the DIANA forecasting platform and forecaster workstations for real-time visualization. Data-sharing mechanisms using APIs are established among BMD, BWDB, and DAE, and forecasting products are published via multiple channels. All stations were integrated into centralized dashboards, and Standard Operating Procedures (SOPs) were developed and implemented. BWDB created a dynamic coastal flood inundation map and automated decision support tools, while DAE established an agromet division with trained staff. Bangladesh's participation in the South Asia Hydromet Forum (SAHF) was institutionalized, with BMD co-chairing the working group on numerical weather prediction.

23. **The outcome was supported by high achievement across intermediate results indicators.** The hydromet network was upgraded beyond the target, reaching 95 percent coverage compared to the planned 60 percent. Sixty-nine meteorologists were trained, surpassing the target of forty. The government approved a post-project operational plan and started its implementation, ensuring long-term sustainability. Regional collaboration targets were exceeded, with 59 staff participating in regional training and three structured dialogues. Approximately four percent of stations experienced physical damage or theft, but these issues were addressed through innovative contract management, improved technical design, and mitigation planning. The combination of systems integration, institutional preparedness, and achieved targets provides strong evidence for a high efficacy rating.

B.2. ASSESSMENT OF ACHIEVEMENT OF THE SECOND OBJECTIVE OF IMPROVING ACCESS TO SUCH SERVICES BY PRIORITY SECTORS AND COMMUNITIES

24. **The outcome of improving access to weather and climate services for priority sectors and communities was achieved and is rated 'Substantial.'** The Project met its second development objective, as measured by two PDO-level results indicators: improved service delivery to agriculture sector beneficiaries and the establishment of a user satisfaction measurement system. Both targets were fully achieved – 80 percent of beneficiaries, including 25 percent of women, were reached, and the satisfaction system was operationalized with endline surveys. However,

the rating is assessed as substantial due to a limited ability to quantify broader impact across categories due to missing clarity on linked intermediate results indicators defining which sectors or communities are targeted.

25. **The project documentation at appraisal was built on the assumption that tailored information products and multi-channel dissemination would enable more timely, informed decision-making by end users.** This outcome rested on the logic that improved access to actionable weather and climate services – especially in hydrology, aviation, and agriculture – will strengthen resilience and productivity among affected communities. To achieve this, the Project emphasized not only generating forecasts but also translating them into locally relevant advisories and ensuring their delivery through a range of accessible platforms. Institutional capacity and user engagement were critical.

26. **The Project supported the development of platforms, tools, and institutional systems for service delivery.** Inputs included the development of the BAMIS portal, a suite of hydrological forecasting tools, and a hydrological and weather service portal by BWDB and BMD. These systems were backed by technical consultancies, digital infrastructure, user interface design, and training for staff. Forecasting models and decision-support systems were customized to user needs in agriculture, water resources, and disaster risk management. DAE established new agromet units and trained extension staff to interpret and disseminate forecasts to farmers and rural institutions.

27. **The outputs included expanded, accessible, and user-oriented weather and climate services**. Agromet advisories are now produced twice weekly and disseminated through multiple channels, including the BAMIS portal, which recorded 9.5 million views between 2019 and April 2025. Additional communication platforms include 487 agromet kiosks, 65 digital display boards, SMS alerts to 30,000 lead farmers (including 25 percent women), and 13 community radio stations that can be received in an area estimated to be populated by up to 18 million people[5]. BWDB launched a decision support system and produced flood forecasts, drought outlooks, and groundwater advisories. BMD made enhanced forecasts publicly accessible through its updated portal. Collectively, these systems represent a substantial increase in the reach and relevance of services nationwide.

28. **All intermediate results indicators related to service access were met or exceeded.** For IRI 5, the Project delivered all targeted outputs: enhanced public weather services by BMD, hydrological advisories by BWDB, and customized agromet advisories by DAE. For IRI 6, agromet information reached 96 percent of districts and upazilas, well above the 70 percent target. The simplified user satisfaction tool was developed and deployed, completing the second PDO indicator. These results are particularly notable given the absence of formal agromet services before the Project.

29. **The outcome was affected by some limitations in measurement and institutional uptake.** The Project met access targets but lacked mechanisms to evaluate user behavioral change or quantify community resilience outcomes. These gaps reduce the depth of outcome evidence but do not undermine the Project's core achievements. In summary, the Project succeeded in laying the foundations for enhanced service delivery across sectors and communities, supporting a substantial rating for Outcome 2.

B.3. JUSTIFICATION OF OVERALL EFFICACY RATING

30. Overall efficacy is rated 'Substantial'. This rating reflects the full achievement of both PDO outcomes. A higher rating would require clearer disaggregation of intermediate results across priority sectors and improved measures of actual development outcomes, such as increased agricultural yield from the agromet service. Although critical sectors and

[5] Estimated based on station locations, population density, and average radio signal reach.

communities benefiting from the Project are identified, more explicit ex-ante definitions would enhance the M&E framework. While the high number of users across channels is proof of the agromet service's utility, a more methodologically sound baseline study would have been needed to justify a high overall efficacy rating.

C. EFFICIENCY

C.1. ASSESSMENT OF EFFICIENCY AND RATING

31. **The project efficiency is rated 'Substantial' based on the rate of return of the investments in weather and climate services in Bangladesh.** The economic analysis is based on the total actual project costs of all components and activities. The economic analysis at appraisal was replicated as much as possible. Project extensions are not assumed to have increased administrative costs significantly as they mostly recovered initial delays with restricted expenses. The underlying assumptions, methodology, and details of the economic efficiency analysis are described in Annex 4. While it is inherently challenging to quantify the actual benefits of hydromet projects, the methodology is based on global best practices and considered sufficiently sound to support a comparison between appraisal and closing and to demonstrate a positive return on investment.

32. **The Economic Internal Rate of Return (EIRR) at completion is estimated at 53.7 percent.** The Net Present Value (NPV) is estimated to be between US$395 and US$1,122 million, and the Benefit-Cost-Ratio (BCR) at completion is estimated to be between 5.3 and 7.8 with the assumption of 12 and 6 percent discount rates respectively. When comparing the estimates using the same methodology at appraisal and closing, the Project outperforms the initial estimation due to cost savings and cancelations (see Annex 4 for detailed comparison). Monte Carlo simulations result in a zero probability of low economic returns for the Project. Based on the recalculated efficiency matrix at appraisal, efficiency at completion shows improvements and leads to a positive efficiency rating.

33. **The implementation efficiency is rated as 'Substantial.'** The Project was implemented through a decentralized structure, with the Project Coordination Unit (PCU) providing overall coordination and three dedicated PIUs at BMD, BWDB, and DAE. While this setup led to delays in coordination, especially in the early stages, due to limited familiarity with World Bank procedures, it allowed for sectoral ownership which is critical for success and sustainability. Over time, implementation efficiency improved significantly. Despite COVID-19-related disruptions and administrative and procurement delays leading to extensions totaling 23 months, the Project disbursed over 91 percent of its allocation and fully achieved its outcomes. Activities were completed by project closing, and handover to government inventories were managed effectively.

D. JUSTIFICATION OF OVERALL OUTCOME RATING

34. **The Project is perceived as successful in achieving its objectives with an overall outcome rating of 'Satisfactory.'** From a broader perspective, hydromet projects – especially the first such operation implemented in a country – are inherently complex and require time to execute effectively (on average, eight years for World Bank hydromet projects) due to the technically complex nature of the subject, involvement of numerous stakeholders, and multiple interdependencies between system components that require a detailed design target to the specific needs of the services, alongside with an organizational and digital transformation. This, combined with the COVID-19 pandemic, changing government priorities that led to an early scope reduction, and the political crisis in 2024 – diverting the focus away across the Project – made the conditions for success challenging. Despite minor opportunities for

improvement listed throughout this ICR, the Project has fully achieved its unchanged PDO of strengthening Bangladesh's capacity to deliver reliable weather, water, and climate information services and improve access for priority sectors and communities, using 63 percent of the initial budget.

E. OTHER OUTCOMES AND IMPACTS

E.1. GENDER

35. **The Project integrated gender considerations into agromet services, ensuring that weather and climate information reached women farmers and decision-makers in agriculture.** Recognizing that women in Bangladesh's agricultural sector often have limited access to extension services and climate advisories, the Project prioritized their inclusion by designing targeted dissemination mechanisms and ensuring that 25 percent of recipients of agromet advisories were female farmers. The gender-disaggregated database of leads of farmer groups allowed targeted and traceable agromet dissemination to women. By leveraging SMS-based advisory systems, local extension networks, and agromet kiosks, the Project increased overall women's engagement with climate-smart agriculture.

E.2. INSTITUTIONAL STRENGTHENING

36. **The Project made significant contributions to institutional strengthening by enhancing the technical, operational, and human resource capacity of key agencies responsible for hydro- and agromet services.** A critical achievement was signing an MoU between all key institutions, fostering structured collaboration and data-sharing agreements between the BMD, BWDB, and DAE. The MoU helps overcome long-standing institutional silos and enables more integrated and coordinated monitoring, forecasting, early warning dissemination, and climate advisory services that last to bear fruits after project closing. Further, MoUs were signed by DAE and university departments, which led to more active knowledge and data exchange and collaboration, resulting in joint publications.[6]

37. **Integrating various systems led by BMD strengthened overall system understanding and ownership.** The Project, especially Component A, was implemented through many packages. PIU and BMD staff were involved in each package as well as the integration of all system components. While challenging, it resulted in a better understanding of the BMD of the entire system compared to firms building and handing over ready-made (sub-) systems.

38. **BWCSRP focused on capacity building, with training programs conducted for meteorologists, hydrologists, and agricultural extension officers to improve technical expertise in weather forecasting, flood modeling, and agrometeorological advisory development.** The Project supported partnerships with universities[7] and research institutions and supported the establishment of new university programs, contributing to a better pipeline of meteorologists and forecasters, as well as the development of a pool of expertise in the country. Regional knowledge exchange through project-supported initiatives allowed Bangladesh to share lessons and adopt best practices from neighboring countries. This solidified institutional capabilities in hydromet service delivery and improved data-sharing channels across meteorological systems.

E.3. MOBILIZING PRIVATE SECTOR FINANCING

[6] Leading to joint publications, for example, Md. Hasan Iman et al. 2024. *Agrometeorological Information for Climate Resilient Agriculture in Bangladesh.* WMO Bulletin, Vol. 73/2, *and* Journal of Agrometeorology; Md. Hasan Iman et al. 2024. *Assessing and Mapping the Vulnerability Index of Bangladesh to Natural and Climate-Induced Disasters: A Spatial Analysis at the Subdistrict Level.* Journal of Agrometeorology, Vol. 26/4.
[7] Listed in component descriptions of Component A and C.

39. **The Project expanded the hydromet sector's engagement with private sector actors, particularly in the procurement, operation, and maintenance of hydromet equipment.** The increasing demand for modern weather and climate services created opportunities for private sector investment in hydromet infrastructure, technology, and data-driven services. The Project stimulated a domestic market for weather instrumentation, forecasting software, and ICT-based early warning systems. This attracted suppliers and service providers to expand operations in Bangladesh. This development could be observed based on the national response of vendors at the beginning compared to the end of the Project, mainly due to performance-based contracts and advanced maintenance clauses, leading to indirect but observable supply capacity improvements.

40. **While the Project did not include a formal Maximizing Finance for Development (MFD) approach, it built the foundation for private sector engagement in climate services by enhancing the availability of high-quality weather and climate data, which can support industries such as agriculture, insurance, logistics, and renewable energy.** While direct private financing was not part of the Project, the improved institutional framework and technical capacity for hydromet services have laid the groundwork for greater public-private collaboration in the sector. Improving hydromet information reliability creates opportunities for firms offering tailored climate services. These are particularly relevant in precision agriculture, disaster risk insurance, and climate-smart advisory solutions.

E.4. POVERTY REDUCTION AND SHARED PROSPERITY

41. **The Project contributed to poverty reduction and shared prosperity by improving access to reliable weather, water, and climate information, particularly for rural and vulnerable populations most affected by climate variability and extreme weather events.** By strengthening EWS and agromet services, the Project helped smallholder farmers, fishers, and disaster-prone communities make more informed decisions about crop planning, water management, and disaster preparedness, reducing losses and enhancing productivity. Improved hydromet services also support national disaster risk management efforts, helping protect livelihoods and assets from climate-related shocks to which the poorest are often most vulnerable.

42. **The Project's focus on agromet information services benefited low-income farming households. This support increased their ability to adapt to climate change and improve agricultural yields.** The Project targeted women farmers and enhanced community-level dissemination of weather advisories. Kiosks, display boards, and community radio stations ensured disadvantaged groups accessed critical information. These interventions support greater economic stability, reduced vulnerability to climate-related income losses, and strengthened resilience among poor and marginalized populations, reinforcing Bangladesh's goals of inclusive development and climate adaptation.

E.5. OTHER UNINTENDED OUTCOMES AND IMPACTS

43. **BMD's upgraded observation network has enabled the country to feed data into global models coordinated by the World Meteorological Organization (WMO)**, allowing Bangladesh to benefit from and contribute to improved global forecasting. The Systematic Observations Financing Facility (SOFF) is now building on this to improve weather forecasts, EWS, and climate information services further.

III. KEY FACTORS AFFECTED IMPLEMENTATION AND OUTCOME

A. KEY FACTORS DURING PREPARATION

44. **Limited readiness for implementation:** At the time of preparation, institutional and technical capacity constraints were evident, particularly regarding procurement, forecasting technologies, and human resource availability. The BMD, which implemented a World Bank-financed project for the first time, had been entrusted with the highest allocations and responsibilities. The lack of preliminary procurement arrangements and readiness assessments caused early delays. While numerous risk mitigation measures were detailed in the PAD Annex 3 (e.g., committees, technical assistance, training) and increased the capacity, additional mitigation measures could have resulted in faster implementation start.

45. **Lack of a dedicated project management structure at inception:** The Project was initiated without fully staffed and operational PIUs, leading to delays in the initial work plan finalization and procurement strategy development.

B. KEY FACTORS DURING IMPLEMENTATION

B.1. FACTORS SUBJECT TO THE CLIENT'S CONTROL

46. **Procurement and contract management:** The Client initially faced challenges managing procurement processes, leading to delays in contracting key equipment, consultancy services, and training programs. Multiple rounds of revisions of procurement documents and limited familiarity with World Bank procedures further contributed to inefficiencies. Two out of the three agencies had not previously implemented World Bank-financed projects. Procurement capacity improved significantly during the implementation, which resulted in the successful completion of procurements and contract management.

47. **Institutional complexity and delays in interagency coordination:** The Project was designed to modernize hydro- and agromet services across multiple agencies, requiring a high degree of inter-institutional coordination. However, the initial design underestimated the challenges of aligning the agencies under a cohesive implementation framework, requiring further adjustments and time to build trust and conditions for effective collaboration. Delays in formalizing institutional collaboration agreements between BMD, BWDB, and DAE resulted in fragmented early implementation. This slowed progress on data-sharing mechanisms, co-location of observation networks, and system integration.

48. **Changing government priorities at early project stages**: At the beginning, the Project experienced delays in decision-making and resource allocation due to shifting priorities within the national government, leading to delays and uncertainty and necessitating an early and substantive restructuring. Another key challenge was paused recruitment and promotion of officials at BMD since 2018 due to a restructuring effort, that led to limited human resources to drive the implementation.

49. **Limited staffing and capacity constraints:** PIUs across implementing agencies faced staffing shortages, with frequent turnover of key personnel. This affected continuity in decision-making, technical oversight, and project monitoring, requiring repeated efforts to onboard and train new staff. Late onboarding of M&E staff led to delayed baseline studies and constraints for effective pre/post comparisons at Project closing.

B.2. FACTORS SUBJECT TO THE WORLD BANK'S CONTROL

50. **Proactive supervision and flexibility:** The Bank played a critical role in addressing implementation challenges through hands-on support in procurement and active technical assistance through the involvement of multiple international, highly experienced hydromet and ICT experts. To minimize procurement delays, the Bank provided

guidance on global best practices and provided training, enabling course corrections to accelerate contract execution. Consultants with technical and procurement expertise ensured hands-on support during critical periods.

51. **Effective knowledge exchange and regional collaboration:** The World Bank facilitated regional knowledge-sharing efforts, connecting Bangladesh with similar hydromet modernization projects in South Asia and facilitated active participation in SAHF. These initiatives helped introduce best practices in forecasting, climate services, and early warning production and dissemination.

B.3. FACTORS OUTSIDE OF THE CONTROL OF CLIENT AND WORLD BANK

52. **Technical complexity of hydromet systems:** The inherent technical complexity of hydromet modernization created implementation challenges that affected project timelines. Hydromet modernization involves multiple interdependent technical systems that must function as an integrated whole, from observation networks to data management systems to specialized forecasting platforms. This technical complexity presented several challenges:

    (a) *Specification Development:* Developing detailed technical specifications for specialized and forecasting software requires significant expertise that is not readily available within implementing agencies.

    (b) *Integration Requirements:* The Project required seamless integration of new systems with existing infrastructure and different agencies' platforms, necessitating careful technical coordination.

    (c) *Sequential Dependencies:* Many implementation activities had strict sequential dependencies, which needed to function correctly before forecasting products could be developed.

    (d) *International Sourcing:* Specialized hydromet equipment required international procurement, introducing additional complexities in import procedures, customs clearance, and supplier coordination.

    These complexities impacted the Project's implementation timeline, particularly during the initial years when agencies were developing specifications, establishing implementation structures, and building capacity. While partial mitigation can be done, the complexity of hydromet projects remains an inherent external factor.

53. **COVID-19 pandemic**: The global pandemic severely disrupted project implementation, causing delays in procurement, supply chains, and travel restrictions that affected technical assistance and capacity-building activities. The pandemic hit at a particularly inconvenient time of project implementation when procurement processes had been finalized, and shipping and installation started, which were severely limited by COVID-19 measures. As a result, this led to delays, insufficient supervision of field operations by international contractors, and more supervision by less experienced local firms, and forced a shift toward virtual training programs, stakeholder engagements, and interactions with limited effectiveness.

54. **Political Crisis in 2024:** The political turmoil in 2024 introduced uncertainties in decision-making and institutional operations, affecting project oversight and the ability of implementing agencies to execute activities, as well as complicating the processing of project duration extension requests, critical for completing the system integration works. The political change close to closure also affected discussions about sustainability measures such as funding, human resources, and signing of operation and maintenance contracts, but all measures were finally implemented.

IV. BANK PERFORMANCE, COMPLIANCE ISSUES, AND RISK TO DEVELOPMENT OUTCOME

A. QUALITY OF MONITORING AND EVALUATION (M&E)

A.1. M&E DESIGN

55. **The Project's M&E framework was designed to track progress toward achieving the PDO through well-defined indicators across hydro- and agromet services.** While the ToC was not spelled out in the PAD (as common at the time of preparation), the connection between activities, outputs, outcomes, and PDO is clear and logical. The PDO indicators captured improvements in forecasting capacity, sectoral service delivery, and user satisfaction, ensuring a balance between technical and impact-based measurements. The text-based PDO-level results indicator permitted capturing multidimensional capacity improvements. Additionally, the inclusion of disaggregated gender data in agromet services demonstrated a commitment to tracking inclusivity. Nevertheless, more quantitative and time-bound indicators, particularly for accuracy improvements and lead times, would have further enhanced the results framework by focusing on sectors and communities in the intermediate results indicators.

A.2. M&E IMPLEMENTATION

56. **The M&E system was progressively strengthened during implementation, with regular data collection, performance reviews, and adjustments based on evolving needs**. Although baseline studies were delayed and data harmonization across BMD, BWDB, and DAE faced challenges, the system improved with digital reporting and real-time monitoring tools. However, delays in finalizing key procurement contracts initially affected the availability of data, and inconsistent sampling between baseline and endline analyses made reliable evaluation challenging.

A.3. M&E UTILIZATION

57. **M&E data from the Project informed decision-making and enabled course corrections.** Regular or real-time insights into the state of the observation networks, including downtimes, allowed for adapting the approach for scaling, maintenance, and contract management. Detailed agency-wise reports for the mid-term review provided a comprehensive overview. The hiring of consultants to prepare the Clients' project completion reports, incl. extensive field data collection, would have been an opportunity to understand the actual impact and use of data, but methodological shortcomings limit the use of final impact assessments.

A.4. JUSTIFICATION OF OVERALL RATING OF QUALITY OF M&E

58. **Overall, M&E quality is rated 'Substantial' because the system effectively tracked progress, informed decision-making, and supported adaptive implementation.** While early challenges in indicator alignment, data gaps, and cross-agency coordination limited initial effectiveness, these were progressively addressed, leading to functioning M&E with few limitations. The results framework and its monitoring permitted progress tracking.

B. ENVIRONMENTAL, SOCIAL, AND FIDUCIARY COMPLIANCE

59. **Environmental Assessment (OP) (BP 4.01), Indigenous Peoples (OP) (BP 4.10), Involuntary Resettlement (OP) (BP 4.12), and Projects on International Waterways (OP) (BP 7.50) were triggered.** Compliance with most of them was rated as 'Satisfactory' during the total project duration. Minor restrictions are detailed below.

60. **Environmental** risks were moderate, primarily linked to the installation of hydromet infrastructure (e.g., weather stations). Compliance with environmental safeguards was maintained, and environmental management plans, including e-waste management plans, were integrated into procurement contracts, incl. regular supervision. The overall safeguards rating and OP 4.01 were downgraded to 'Moderately Satisfactory' between the Implementation Status and Results Reports (ISR) 7 and 14. This change was due to the pending disclosure of screening sheets, non-

compliance with EMP implementation for three work packages, inadequate due diligence on physical works, and a vacant environmental specialist position. The Bank emphasized improving documentation, internal monitoring, and compliance, and a safeguard clinic was conducted to strengthen PIU capacity. ISR 14 and 15 confirm that all corrective action items have been completed, all monitoring progress reports were submitted on time, and the overall implementation performance is satisfactory. E-waste management would have benefitted from enhanced capacity building and exposure visits. Still, overall monitoring and implementation of the EMP and other safeguard documents and a committed project team ensured that any emerging issues were addressed.

61. **Social** compliance focused on ensuring inclusive access to climate services, particularly for women farmers and vulnerable communities. The Project successfully integrated gender-disaggregated tracking of data. It ensured that agromet advisories reached 25 percent of women beneficiaries through SMS notifications to lead farmers. No significant social risks were reported, and no concerns for labor influx were identified in any of the three components, as most activities are related to rehabilitation or refurbishment. Several local Grievance Redress Committee (GRC) meetings and training sessions were conducted. All sites were screened, and the screening report confirmed no that there were no impacts on land acquisition, resettlement, or Indigenous Peoples. The Bank's E&S team provided four training sessions to all IAs on screening, grievance management, and recording, code of conduct, occupational health and safety, and COVID-19 protocols, while the IAs conducted a national safeguard workshop. IAs addressed all 14 grievances. The social performance was satisfactory throughout the project period.

62. **Financial management** performance was generally satisfactory, with timely audits and adherence to financial reporting requirements. However, the Project experienced initial challenges in staffing, fund disbursement, and financial tracking due to delays in procurement approvals and slow budget execution in the early years. These issues were progressively resolved with capacity-building efforts and stronger oversight. At 10.2 percent of the total cost, Borrower financing was consistent with the anticipated 11.6 percent at appraisal.

63. **Procurement** delays were one of the most significant challenges during project implementation. The complexity of hydromet equipment procurement, international vendor engagement, and technical specifications led to multiple revisions and delays, further complicated by COVID-19. Some contracts required rebidding due to non-compliance with specifications, causing further disruptions. Contract management issues such as timely processing of amendments arose. Despite these challenges, procurement was completed satisfactorily. Significant support from the World Bank's technical team streamlined procedures and ensured timely execution.

C. BANK PERFORMANCE

C.1. QUALITY AT ENTRY

64. **The Bank played a key role in project design, leveraging its experience in hydromet modernization to structure a project that addressed Bangladesh's critical forecasting and early warning gaps.** The World Bank designed the Project as part of a broader regional strategy to enhance resilience to weather- and climate-related hazards in South Asia, using learned lessons from other countries in the region. At the time of preparation, Bangladesh had limited capacity for accurate forecasting, with outdated and mostly manual systems, weak data-sharing infrastructure, and insufficient staff at involved agencies. The Bank designed the Project effectively to fill these gaps.

65. **The Bank structured the Project around national priorities but with a regional dimension, meeting the criteria for regional IDA financing.** It was part of a sequenced approach in South Asia, complementing similar operations in

Nepal and Bhutan. The Bank plays a strategic role in providing technical and institutional support at the regional level, and while a consolidated regional program is challenging to implement, strengthening knowledge platforms and facilitating experience-sharing and collaboration among task teams and project counterparts, as also done under the Project is deemed highly beneficial for regional hydromet services. The Bank's choice of the IPF instrument was appropriate, allowing for infrastructure investments and institutional capacity strengthening. The design featured four components targeting hydromet systems modernization, agromet service development, and contingency emergency response, with embedded technical assistance and training across all implementing agencies, constituting a suitable project design developed by the Bank at entry. The initial Financing Agreement was not fully aligned with the PAD and missed civil works as eligible expenses. However, this was corrected through the first restructuring.

66. **The Bank built on global and regional lessons to design the Project with emphasis on the importance of user-responsive services, particularly in agriculture, disaster risk management, and flood forecasting.** The Bank's design introduced a Joint Technical Working Group and required MoUs among implementing agencies to formalize data sharing and coordination. Recognizing limited in-country expertise, the Project included support from two systems integrators (supporting BMD and BWDB, respectively) and specialized agromet expertise.

67. **Capacity constraints, including the absence of university programs in meteorology, were anticipated by the Bank's team and addressed through dedicated training and institutional strengthening plans.** While the Project was ambitious and well-aligned with government and regional objectives, the design had some shortcomings – particularly in its results framework, which did not fully capture the sectoral improvements. Nonetheless, the Project's scope, implementation arrangements, and risk mitigation measures were well thought out, and the World Bank team's performance at entry is rated *Satisfactory*.

C.2. QUALITY OF SUPERVISION

68. **The World Bank provided close technical support throughout the project implementation, with 16 documented implementation support missions covering procurement progress, mid-term review, restructuring approvals, project completion assessments, and regular and ongoing technical missions.** All ISRs, Aide Memoires, and Management Letters were filed timely and describe progress, constraints, and actions taken to resolve challenges in a detailed and neutral manner, covering project-level issues as well as environmental and social, and fiduciary aspects. The Bank's technical experts, including multiple local and highly specialized international consultants, maintained high-level support and regular field visits to work through technical challenges with the Client. This hands-on engagement was critical in resolving delays related to procurement, contract execution, and system integration. International hydromet expertise ensured the alignment of the Project with global best practices and enabled regional knowledge and use of existing platforms and networks.

69. **The Bank proactively addressed procurement and institutional challenges by providing detailed guidance on hydromet equipment procurement packages.** Technical specialists ensured that specifications matched context-specific operational needs, helping to mitigate procurement challenges. Following procurement, proactive project management consisting of hands-on support, capacity development at implementing agencies and vendors, and frequent technical missions helped to ensure effective contract management, that installations complied with specifications in the bidding documents, and that vendors adhered to stated requirements. The World Bank

leveraged global knowledge from across its hydromet portfolio with a focus on South Asia/India, helping Bangladesh to adopt best practices. This engagement strengthened institutional learning, technical competency, and sustainability, ensuring the Project's impact extended beyond immediate outputs.

C.3. JUSTIFICATION OF OVERALL RATING OF BANK PERFORMANCE

70. **Bank Performance is rated 'Satisfactory' due to strong technical engagement, problem-solving support, and proactive supervision.** While procurement and capacity risks could have been better anticipated in preparation, the World Bank's adaptive approach and technical guidance played a critical role in overcoming challenges and transforming a temporary problem project into one with highly successful outcomes.

D. RISK TO DEVELOPMENT OUTCOME

71. **The risk to sustaining project outcomes is 'Substantial,' mitigated through the institutional strengthening efforts and formalized ownership transfer of hydromet equipment and services.** The Project built long-term capacity within BMD, BWDB, and DAE, improving the pipeline of trained professionals and strengthening the integration of forecasting into decision-making processes. The Project ensured adequate handover of systems to the respective departments. Through annual maintenance contracts for observation networks and high-end ICT equipment, sustainability is ensured for the following years, which could either be continued long-term or result in a gradual handover to internal maintenance by government departments. Although maintenance contracts are signed, financial sustainability remains a risk for hydromet projects due to the need for ongoing investment. However, in Bangladesh, all sustainability measures were implemented successfully. The operation and maintenance budget has been allocated to the implementing agencies and remains available, thus leading to a substantial but well-anticipated and mitigated sustainability risk.

V. LESSONS AND RECOMMENDATIONS

72. **Many lessons from this Project are common for World Bank-financed hydromet projects[8] and were integrated in the design if already available at the preparation stage, while others were identified after this project started.** These include the importance of proactive engagement with local populations (P006051), advanced forecasting and early warning systems (P006051), and centralized and reliable hydromet system that can provide crucial weather insights for multiple sectors, from targeted agriculture to disaster response (P009549). Other known lessons covered in the World Bank's 2023 report on hydromet services[9] touch on the challenge of sustainability of operations (P006051, P009549), sector-specific capacity building needs (P009467, P009549), difficulties with infrastructure maintenance costs (P009549), hydromet systems' need for adaptation to local conditions (P006051), and the need for comprehensive planning for long-term benefits (P009468).

73. **To add to this existing knowledge, the ICR focuses only on new or refined lessons learned.** An extended list with equally relevant recommendations that, however, target foremost technical task teams preparing other hydromet projects can be found in Appendix 2.

[8] Based on a review of hydromet operations in database of lessons learned of the Independent Evaluation Group (IEG).
[9] Grimes, D. R., et al. 2022. *Charting a Course for Sustainable Hydrological and Meteorological Observation Networks in Developing Countries.* Washington, DC.: The World Bank. http://hdl.handle.net/10986/38071.

74. **Connecting hydromet services to a sector application like agriculture ensures that improved forecasts reach end-users and deliver measurable development outcomes while directly showing the Project's impact.**

   (a) *Context:* By integrating agromet services into its design, the Project successfully showed how improved forecasts reach the end-users, in this case farmers, translate into actionable insights for agricultural planning, enhancing productivity and resilience. This permits development outcome reporting *during* the implementation period.

   (b) *Recommendation:* Future hydromet modernization projects should systematically incorporate tailored service components that demonstrate the value of improved data in priority sectors, such as early warning. This approach ensures that investments in data collection and forecasting lead to real-world applications. Strong partnerships with sectoral agencies should be established to align service design with user needs, ensuring sustainable co-production processes responding to evolving user needs and service opportunities.

75. **Formalizing inter-agency cooperation early through agreements and shared programs enhances data sharing, institutional capacity development, and long-term service sustainability.**

   (a) *Context:* Institutional strengthening was a key success factor in the Project, particularly through formalized collaboration among BMD, BWDB, and DAE. The signing of a *Memorandum of Understanding (MoU)* facilitated data sharing, while engagement with academic institutions supported the creation of specialized degree programs and research initiatives. Additionally, regional partnerships under SAHF strengthened these partnerships and provided additional avenues for knowledge exchange.

   (b) *Recommendation:* Future projects should ensure that institutional coordination mechanisms are formalized early through binding agreements such as MoUs. Capacity-building programs should be embedded within project frameworks, leveraging partnerships with academia to enhance technical expertise in weather, water, and climate services. Regional collaboration platforms should be expanded to facilitate cross-border knowledge sharing and operational sustainability.

76. **Specialized and intensive hands-on extended implementation support is critical for successful hydromet modernization projects in developing countries.**

   (a) *Context:* The Project benefited significantly from hands-on technical support provided by the World Bank team, particularly in addressing complex procurement challenges, system integration issues, and institutional coordination. Hydromet systems' technical complexity demanded more intensive support than typical projects, especially as it was the first World Bank-funded project for some agencies during the critical early stages of specification and procurement.

   (b) *Recommendation:* Early and ongoing hands-on technical support is critical for hydromet projects, including: (i) dedicated technical advisors working with implementation units; (ii) specialized procurement clinics and technical workshops during the project launch phase; (iii) frequent technical missions during critical implementation periods; and (iv) systematic technical reviews at key milestones. The approach should be incorporated into project design and budgeting, with adequate resources for specialized technical assistance throughout the implementation. This enhanced support model would help address the common implementation challenges in hydromet projects and increase the likelihood of successful and timely delivery.

77. **Engaging academic institutions and enabling regional knowledge exchange helps build future capacity, supports implementation, and strengthens the hydromet workforce and sector through training and joint research.**

    (a) *Context:* The Project successfully demonstrated the benefits of integrating academic institutions into hydromet modernization efforts and led to active participation at regional events such as SAHF. In Bangladesh, new academic graduate programs were established. Universities contributed technical expertise, enabling students to support project activities. In return, students gained valuable hands-on experience, enhancing their employability and strengthening the sector's workforce.

    (b) *Recommendation:* Future projects should include participation at regional platforms and establish formal internship and research collaboration programs with universities, including long-term agreements for knowledge exchange. Paid internships could be institutionalized within hydromet agencies, offering competitive compensation to attract top talent. Hydromet agencies should create pathways for interns to transition into permanent roles. This ensures a continuous supply of trained professionals. Joint research initiatives, co-publications, and integrating hydromet topics into university curricula should also be promoted.

78. **Building trust in new hydromet systems requires parallel testing with legacy systems. It also demands transparent validation and clear communication of benefits to users.**

    (a) *Context:* Introducing modern hydromet systems often faces resistance due to concerns about. The Project mitigated this by operating new systems alongside existing established and manual processes, with verification and validation processes in place to compare performance, enable any required adjustments, and create trust in the upgrades. For example, flood forecasting models were tested in parallel with legacy systems during the 2024 monsoon season, allowing forecasters to validate outputs before full adoption.

    (b) *Recommendation:* Future projects should plan and reserve sufficient time for structured parallel testing phases when introducing new technology. Automated station data should be cross-checked with manual observations to build confidence among users, including local maintenance and analog weather monitoring teams. Agencies should also establish clear plans to ensure complete transition to new systems while ensuring key personnel receive adequate training on new technologies. Data service models can be considered, while institutional capacity in data quality assurance and quality control should be strengthened. Communicating the reliability and accuracy of modernized systems to end-users is essential for fostering trust and ensuring successful adoption.

79. **The sustainable operation of hydromet infrastructure depends on forward-looking procurement strategies, including maintenance planning and funding integration.**

    (a) *Context:* The project highlighted the importance of ensuring the long-term sustainability of newly installed observation networks. Many hydromet systems fail post-project due to lacking maintenance funding, trained personnel, and structured operational support. Agencies often face challenges sustaining critical infrastructure. These challenges lead to equipment failures and service disruptions.

    (b) *Recommendation:* Future projects should further shift toward institutional integration by embedding hydromet systems into core operational frameworks. This includes pre-negotiating (during initial procurement) and signing annual maintenance contracts before the warranty period ends – mainstreamed in regular government budgets, implementing maintenance tracking software for automated issue reporting, and ensuring spare parts procurement strategies are established.

ANNEX 1. RESULTS FRAMEWORK AND KEY OUTPUTS

A. RESULTS FRAMEWORK

PDO Indicators by Outcomes

Strengthen GoB's capacity to deliver reliable weather, water and climate information services								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
PDO1: Improved capacity for forecasting (Text)	No verification; 24 forecasts for public weather service	Jun/2018	25% Increase in skill over baseline	Dec/2022	Improvement in forecasts by at least one attribute (skill/lead time/ frequency/spatial range)	Nov/2024	Improvement in forecasts by at least one attribute achieved	Nov/2024
	Comments on achieving targets		Both BMD and BWDB improved their forecasting capacity (skill, lead time, frequency and/or spatial range) and have finalized notes presenting the relevant evidence.					

Improve access to services by priority sectors and communities								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
PDO2: Improved service delivery to targeted agriculture sector beneficiaries (Percentage)	0.00	Jun/2017	70	Dec/2022	80.00	Nov/2024	80.00	Nov/2024
	Comments on achieving targets		30,000 identified lead farmers from 15,000 Farmers' groups are receiving agromet information including special advisories through short message service (SMS) and IVR (Voice message) of which Female is 25%. DAE is in the process of checking the current list and expanding the list further to include key members of the community. BAMIS portal has been established and has had over 7.7 million views since it went online in 2019. Agromet services are being developed and delivered regularly as National, District and Special Bulletins.					
of which Female are (Percentage)	0.00		40		25.00		25.00	
	No	Jun/2017	Survey is conducted	Dec/2022	Yes	Nov/2024	Yes	Nov/2024

PDO 3: Systematic measurement of user satisfaction in place, to measure improvements in weather information service delivery: Establishment of an user satisfaction measurement system (Yes/No)	Comments on achieving targets	For all three components baseline, midline and end-line studies have been completed and reports prepared. Furthermore, simplified survey tools were developed for all three PIUs. In addition, DAE and BWDB implemented pop-up windows on their website to continuously receive user feedback.

Intermediate Indicators by Components

Component A- Strengthening Meteorological Information Services								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
IR1: Upgraded and Modernized Hydromet related Network Established (Percentage)	0.00	Jun/2017	70	Dec/2022	60.00	Nov/2024	95.00	Nov/2024
	Comments on achieving targets		BMD: - Synoptic AWS: 35 installed, 35 functioning - AWOS: 3 installed, 3 functioning - Ag AWS: 124 installed, 120 functioning, 3 stolen, 1 damaged by flood/washed away/human factor - ARG: 65 installed, 59 functioning, 6 removed - Hydrogen gas generator: 11 installed, 6 functioning, 5 not functioning BWDB: - Climate stations: 3 installed, 3 functioning - ARG: 272 installed, 270 functioning, 1 not functioning, 1 damaged by flood/washed away/human factor - Ground water: 905 installed, 866 functioning, 6 stolen, 18 damaged by flood/washed away/human factor, 5 removed - Surface water monitoring stations: 315 installed, 296 functioning, 1 not functioning, 7 stolen, 6 damaged by flood/washed away/human factor), 5 removed - Coastal water level: 40 installed, 32 functioning, 2 not functioning, 1 stolen, 5 damaged by flood/washed away/human factor DAE: - Ag-AWS: 160 installed, 152 functioning, 7 stolen, 1 damaged by flood/washed away/human factor						
IR2: Strengthened Pipeline of Meteorologists in the Country (Number)	0.00	Jun/2017	75% of targeted training completed	Dec/2022	40.00	Nov/2024	69.00	Nov/2024
	Comments on achieving targets		To date, 69 students finalized a master’s program in meteorology of which 35 are female. Currently 11 students are continuing the master's program of which 6 are female. Department of Meteorology at Dhaka University started in (2022) the Bachelor program and currently 76 students of which 40 are female are enrolled. Furthermore, under component C, BAU and BSMRAU have initiated master’s programs in agro-meteorology with a current enrollment of 19 students (8 female) and 11 students (7 female), respectively.						
IR 3: Improved Financial Sustainability to Maintain Hydromet Networks (Text)	No operational plan exists	Jun/2017	Post project operational plan approved	Dec/2022	Post project operational plan approved, and	Nov/2024	Post project operational plan approved, and	Nov/2024

Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
					implementation initiated		implementation initiated	
	Comments on achieving targets		The post-project operational plans include different aspects critical for sustainability: For all three agencies, warranty and maintenance were negotiated together with the purchase and installation of the observation networks to provide continued support with operation and maintenance; standard operating procedures were established for operation and maintenance of stations and is available at the stations; many stations were installed on the premises of partner/research institutes which now receive direct access to the data and nominated a focal point which was trained to check and clean the station, as well as report any issues; all three agencies requested additional budget for operation and maintenance of the modernized network.					
Component B- Strengthening Hydrological Information Services and Early Warning Systems								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
IR 4: Improved Regional Dialogue and Exchange on Weather, Water and Climate Services (Text)	-	Jun/2017	Joint partnerships for regional product development strengthened	Dec/2022	Implementing agency participants engage in at least 3 regional dialogue meetings; 10 staff participate in regional trainings	Nov/2024	Implemented agency participants engaged in 3 regional dialogue meetings; 59 staff participated in regional trainings.	Nov/2024
	Comments on achieving targets		Implementing agency participants engaged in 3 SAHF main events, SASCOF and weekly SAHF Forecaster' forum. 7 staff from implementing agencies participated in regional trainings. A joint learning event between Bangladesh and India took place Oct 30 – Nov 5, 2022, involving 11 persons from BMD, BWDB and DAE. A second joint learning event between Bangladesh, Sri Lanka and India took place May 2-8, 2023, involving 11 persons from BMD, BWDB and DAE. Bangladesh participated in two IBF trainings organized by RIMES in the context of SAHF: online preparatory training - virtual (13 persons), basic level training in February 2023 - Dhaka (13 persons). Representatives from all three agencies participated in the SAHF IV conference from February 5-11, 2024 (4 persons).					
IR 5: Increase in Development of Weather and Climate Products. (Text)	Limited or no product available	Jun/2017	Customized Agro advisories; Improved PWS; QPF; QPE; Nowcasting	Dec/2022	BMD: portal for improved Public weather services dissemination established; BWDB: water resource advisories and hydrological	Nov/2024	BMD: A portal for improved public weather services dissemination is under development. BWDB: Water resource advisories and hydrological	Nov/2024

					outlooks; DAE: customized agro-advisories		outlooks under development DAE: Customized agro-advisories	
	Comments on achieving targets		BMD: Enhanced forecasts are available on BMD's newly developed portal. BWDB: Flood Forecasts model expansion and decision support system to produce improved hydrological outlooks and flood inundation mapping, drought and available on BWDB's portal. Ground water forecast/outlook are developed. DAE: Agromet Advisories are developed and disseminated successfully through the BAMIS portal and other communication mechanisms.					
Component C- Agro-Meteorological Information Systems Development								
Indicator Name	Baseline		Closing Period (Original)		Closing Period (Current)		Actual Achieved at Completion	
	Result	Month/Year	Result	Month/Year	Result	Month/Year	Result	Month/Year
IR 6: Improved Access to Agromet Information at the District and Upazila Level (Percentage)	0.00	Jun/2017	50	Dec/2022	70.00	Nov/2024	96.00	Nov/2024
	Comments on achieving targets		BAMIS portal was established in 2019 to generate and disseminate agromet advisories. Agromet kiosks have been installed in 487 Upazilas (out of 492) and are being operationalized to disseminate agromet advisories. 65 digital display boards have been successfully operationalized for dissemination of agromet advisories. 12 community radios stations have been operationalized and provide agromet advisories five days a week for one hour to local farmers, facilitating exchange of information among DAE extensionists and farmers. All dissemination mechanisms receive agromet advisories through BAMIS portal.					

B. KEY OUTPUTS

Objective 1: Strengthen capacity to deliver reliable weather, water, and climate information services	
PDO Indicator	PDO1: Improved capacity for forecasting
Intermediate Results Indicators	IRI1: Upgraded and Modernized Hydromet related Network Established IRI2: Strengthened Pipeline of Meteorologists in the Country IRI3: Improved Financial Sustainability to Maintain Hydromet Networks IRI4: Improved Regional Dialogue and Exchange on Weather, Water, and Climate Services
Key Outputs	1. One thousand nine hundred twenty-one observation stations and accompanying SOPs for operation and maintenance, incl. 905 automatic groundwater stations, 315 river water level monitoring stations, 337 rain gauges, 38 AWS, 283 Ag-AWS (with 50 soil moisture sensors), 40 coastal monitoring stations, and 3 AWOS at Chittagong, Dhaka, and Sylhet international airports. 2. Fifteen single-beam echo sounders, one side-scan sonar, eight total stations, two sub-bottom profilers, 26 portable depth sonars, six drilling machines for groundwater well construction, four survey boats, four catamarans, 18 Acoustic Doppler Current Profilers (ADCPs), 18 Differential Global Positioning Systems (DGPS), 15 Real-Time Kinematic (RTK) units, 42 handheld Global Positioning System (GPS) devices, 36 leveling machines, and various manual monitoring instruments. 3. Upgraded laboratories with sample-based water quality monitoring equipment and installed 20 sondes for automatic water quality monitoring. 4. Data dashboard for acquisition and visualization, as well as maintenance tracking software. 5. HPC for enhanced modeling and system integrating various data sources 6. Improved quality and spatial resolution (18 to 3 km) of forecasts, quadrupled modeling frequency, and enhanced forecast lead time from one to three and outlook from three to five days. 7. Flood forecasting cover more than doubled with 64 districts, and 1D-2D coupled flood inundation forecasting for coastal areas resulting in dynamic flood inundation maps. 8. Sixty-nine master’s degree graduates in meteorology, of which 35 are female. Seventy-six students, of which 40 are female, are currently enrolled in the new bachelor’s program at the Department of Meteorology at Dhaka

	University. BAU and BSMRAU have initiated master's programs in agro-meteorology with a current enrollment of 19 students (8 female) and 11 students (7 female), respectively. 9. Post-project operational plans and annual maintenance contracts are approved, budgeted, and implemented. 10. Participation at three regional dialogue meetings, and 59 staff members participated in regional training activities.
Objective 2: Improve access to such services by priority sectors and communities	
PDO Indicators	PDO2: Improved service delivery to targeted agriculture sector beneficiaries
	PDO3: Systematic measurement of user satisfaction in place, to measure improvements in weather information service delivery: Establishment of an user satisfaction measurement system
Intermediate Results Indicators	IRI5: Increase in Development of Weather and Climate Products IRI6: Improved Access to Agromet Information at the District and Upazila Level
Key Outputs	1. Enhanced meteorological forecasts are available on BMD's newly developed portal. 2. The flood forecast model was expanded, a decision support system was established to produce improved hydrological outlooks, and flood inundation mapping is available on BWDB's portal. 3. Groundwater and drought forecasts and outlooks are developed. 4. Agromet advisories are developed, as well as special agromet advisory services for livestock, poultry, and fisheries. 5. The BAMIS portal (https://www.bamis.gov.bd/) was developed and has recorded approximately 9.5 million views between 2019 and April 2025. 6. Four hundred eighty-seven agromet kiosks, 65 digital display boards, SMS notifications sent to 30,000 lead farmers – including female farmers – as well as 13 community radio stations that are broadcasting agromet information were established. 7. For all three components, baseline, midline, and endline studies have been completed and reports prepared. Furthermore, simplified survey tools were developed for all three PIUs. In addition, DAE and BWDB implemented pop-up windows on their website to continuously receive user feedback.

ANNEX 2. BANK LENDING AND IMPLEMENTATION SUPPORT/SUPERVISION

A. TASK TEAM MEMBERS[10]

Name	Title	Role
Melanie Simone Kappes	**Disaster Risk Management (DRM) Specialist (ADM)**	**TTL [C]**
Mohammad Rafiqul Islam	**DRM Specialist**	**TTL [C]**
Arati Belle	Sr. DRM Specialist (ADM)	TTL [I]
Ignacio M. Urratia Duarte	Sr. DRM Specialist	TTL [I]
Shahpar Selim	Environmental Specialist	TTL [I]
Nadia Sharmin	Sr. Environmental Specialist	TTL [A]
Poonam Pillai	Sr. Urban Development Specialist (ADM)	TTL [A]
Mohammed Atikuzzaman	**Sr. Financial Management Specialist (ADM)**	**FS [C]**
Aisha Lanette N. De Guzman	Sr. Financial Management Specialist	FS [I]
Hasib Ehsan Chowdhury	Sr. Financial Management Specialist	FS [I]
Mohammad Reaz Uddin Chowdhury	Sr. Financial Management Specialist	FS [A]
Mohammed Kamruzzaman	**Sr. Procurement Specialist (ADM)**	**PS [C]**
Mir Mehbubur Rahman	**Procurement Consultant**	**PT-e [C]**
Shourov Kumar Sharma	**Program Assistant**	**PT [C]**
Mahmuda Nusrat Hussain	Sr. HR Assistant	PT-e [I]
Yogesh Bom Malla	–	PS [I]
Ishtiak Siddique	Sr. Procurement Specialist	PS [A]
Bushra Nishat	**Environmental Specialist (ADM)**	**ES [C]**
Sharlin Hossain	**Environmental Consultant**	**ES-e [C]**
Lisbet Kugler	Sr. Environmental Specialist	ES [I]
Md Istiak Sobhan	Sr. Environmental Specialist	ES [I]
S. M. Zulkernine	Sr. Environmental Specialist	ES [I]
Suiko Yoshijima	Sr. Environmental Specialist	ES [I]
Shakil Ahmed Ferdausi	Sr. Environmental Specialist	ES [A]
Sabah Moyeen	**Sr. Social Development Specialist (ADM)**	**SS [A, C]**
Shariful Islam	**Social Specialist**	**SS [C]**
Sabah Moyeen	Sr. Social Development Specialist	SS [A]

[10] For each person, the last (known) title and most relevant role are listed. TTL refers to Task Team Lead, FS to Financial Management Specialist, PS to Procurement Specialist [PT = Procurement Team], ES to Environmental Specialist, SS to Social Specialist, SC to Senior Counsel, FO to Finance Office, TM to Team Member, ETM to Extended Team Member, and (E)TM-e to an external (E)TM. [A] refers to a role held at appraisal, [I] to only implementation (neither A nor C), and **[C] at closing (bolded)**. The table is based on data reported in PAD and ISRs and the World Bank's people database.

Vidya Venugopal	Senior Counsel	SC [I]
Jorge Luis Alva-Luperdi	Senior Counsel	SC [A]
Hisham A. Abdo Kahin	Manager	SC [A]
Satish Kumar Shivakumar	Sr. Finance Officer	FO [A]
Alice Soares dos Santos	**Senior Hydromet Advisor/Expert Consultant**	**TM-e [C]**
Amani Haque	Program Assistant	TM [I]
Angie Harney	Program Assistant	TM [A]
Anish Kumar Bansal	**Hydrologist and Instrumentation Expert Consultant**	**TM-e [C]**
Diego Juan Rodriguez	Lead Economist	TM [A]
Guillermo Siercke	Sr. DRM Specialist	TM [A]
Haris Khan	Sr. DRM Specialist	TM [I]
Iguniwari Thomas Ekeu-Wei	DRM Specialist	TM [I]
Kazi Shahazada Shahanewaz Hossain	**Consultant**	**TM-e [C]**
Luis Corrales	Consultant	TM-e [A]
Makoto Sawa	Sr. DRM Specialist	TM [A]
Marie Florence Elvie	**Program Assistant**	**TM [A, C]**
Md Abul Fayez Khan	Program Assitant	TM [I]
Mohammad Sayeed	Consultant	TM-e [A]
Syed Ahmed Ali	–	TM [A]
Tjark Gall	**Urban Resilience Specialist**	**TM [C]**
Vladimir Tsirkunov	Lead Hydromet Specialist / Consultant	TM [I]
Yunziyi Lang	DRM Specialist	TM [I]
Guna Paudyal	Hydromet Expert Consultant	ETM-e [I]
Jeff Lazo	Sr. Economist / Hydromet Consultant	ETM-e [A]
Laxman Singh Rathore	**Lead Consultant**	**ETM-e [C]**
Mannava Sivakumar	Agromet Expert Consultant	ETM-e [A]
Mark Heggli	Hydromet Expert Consultant	ETM-e [A]

B. STAFF TIME & COST

Stage of Project Cycle	Staff Time & Cost	
	No. of Staff Weeks	US$ (including travel and consultant costs)
Preparation		

FY14	1.325	5,758.98
FY15	25.005	111,669.95
FY16	51.089	230,808.16
FY17	0.000	57,423.52
FY18	0.000	29,034.52
Total	77.42	434,695.13
Supervision/ICR		
FY17	17.583	207,308.59
FY18	36.432	161,823.10
FY19	46.216	244,987.80
FY20	47.157	283,179.05
FY21	55.523	331,789.03
FY22	33.584	153,448.79
FY23	36.644	159,458.82
FY24	30.617	140,185.37
FY25	26.640	136,114.01
Total	330.40	1,818,294.56

ANNEX 3. PROJECT COST BY COMPONENT

TABLE 2: WORLD BANK CONTRIBUTION TO PROJECT COSTS AFTER RESTRUCTURINGS BY COMPONENT (US$ MILLION)

Project Components	Appraisal	2. Restr. (2020)	4. Restr. (2023)	6. Restr. (2024)
A Strengthening Meteorological Information Services	58.9	28.3	25.5	22.2
B Strengthening Hydrological Information Services and Early Warning Systems	40.6	33.6	33.6	33.4
C Agro-meteorological Information Systems Development	13.5	22.5	22.5	19.6
D Contingent Emergency Response	0.0	0.0	0.0	0.0
Total	**113.0**	**84.3**	**81.5**	**75.1**

TABLE 3: PROJECT COSTS BY COMPONENTS AT APPROVAL AND COMPLETION (US$ MILLION)

	At Approval			**Actual**			
Components	World Bank Contribution	Borrower Contribution	Total at Appraisal	World Bank Contribution	Borrower Contribution	Actual Total	Percentage of actual to original
A	58.9	8.6	67.5	23.4	2.2	25.6	38%
B	40.6	4.7	45.3	31.3	4.3	35.7	79%
C	13.5	2.6	16.1	20.6	1.7	22.3	138%
D	0.0	0.0	0.0	0.0	0.0	0.0	0%
Total	**113.0**	**15.9**	**128.9**	**75.3**	**8.2**	**83.5**	**65%**

ANNEX 4. ECONOMIC EFFICIENCY ANALYSIS

1. An economic analysis was performed to assess the return rate of capital investments in hydro-met services in Bangladesh. The study used all the Project's components comprised of 100 percent of the total actual project costs. The following assumptions were made:
   (a) The economic life of the infrastructure built under the Project was 100 years based on the assumption at the appraisal.
   (b) The discount rate of 12 percent was used at appraisal. Based on the updated methodology, the range of discount rates from 6 to 12 percent was used for this analysis.
   (c) The operation and maintenance costs after the project completion were 10 percent of the project costs based on the assumption at the appraisal.

METHODOLOGY

2. **The ICR methodology is tied to the approach used in the Project Appraisal Document (PAD).** The PAD used sensitivity analysis to consider the coarse-grained benefits calculations. This analysis uses Monte Carlo simulations to achieve the same objective. The current study follows the willingness to pay (WTP) and the expected value of statistical lives (VoSL) saved undertaken at the appraisal stage with updated assumptions.
3. **Expected WTP and VOSL were used to calculate EIRR, NPV, and BCR for the project's actual costs and benefits.** Note the three key differences between the PAD and the calculations used for the final economic analysis:
   (a) The appraisal assumed that project investments would incur 10 percent depreciation each year from 2017 to 2022. Current standards of economic efficiency analysis do not use depreciation during the project completion. Thus, this assumption was dropped for the final analysis.
   (b) Project costs are divided by the duration between 2017 and 2022 based on available information from the appraisal. The final analysis uses actual costs from each year between 2017 and 2024.
   (c) It was assumed at the appraisal that component benefits would start accruing to the beneficiaries from the sixth year, the year after the Project was scheduled to be completed. The final analysis assumed all the benefits start accruing after the actual project completion year of 2024.
4. **Based on the descriptive methodology, the discount rate is usually set at 12 percent for the economic analysis of World Bank projects.** A technical note recommends a different approach of calculating the discount rate as the product of the per capita growth rate of the country and the elasticity of marginal utility of consumption.[11] Bangladesh's per capita annual growth rate is expected to be 6 percent in the next 20 years. The elasticity of marginal utility of consumption is harder to calculate but is likely to be in the range of 1 to 2. Therefore, we use the range of 6 to 12 percent discount rate for Bangladesh for the economic analysis at project completion.
5. **Following the PAD, three types of quantitative benefits were included in the final analysis:**
   (a) The value of general improvements in weather information was calculated using (WTP) studies.
   (b) The value of hydro-met services to the agricultural sector was calculated using WTP studies.

[11] The World Bank. 2016. *Discounting Costs and Benefits in Economic Analysis of World Bank Projects*, OPSPQ, May 9, Washington, D.C.

(c) The value of early warning systems was calculated using the expected number of lives saved and the corresponding value of statistical lives (VoSL).

6. **The final analysis used the actual total expenditures at project closing, including the World Bank's and the borrower's contribution for each year as shown in Table 4.**

TABLE 4: TOTAL COSTS OF THE PROJECT, INCLUDING BORROWER'S CONTRIBUTION (IN MILLION US$)

	2017	2018	2019	2020	2021	2022	2023	2024	Total
Total Costs	1.5	11.8	8.7	6.6	23.8	13.9	10.5	4.2	81

7. **It was attempted to replicate the economic efficiency matrix calculated at the appraisal.** Annex 6 of the PAD contained all information necessary to replicate the economic analysis at appraisal, and the original assumptions were retained. The expected annual benefits and annual costs used in the PAD were successfully recalculated. However, the calculated EIRR, NPV, and BCR did not match those reported in the PAD. The differences between the original and the recalculated efficiency matrix are in the top part of Table 5. The only likely explanation is that the formulas used for calculating EIRR, NPV, and BCR in the PAD are different from those currently used.

8. **A Monte Carlo simulation was conducted to consider the likely differences between the actual benefits and the calculated benefits.** A range of ±20 percent differences in the benefits calculations were considered in the simulation. The second part of Table 5 shows the efficiency matrix at completion, along with the simulation results.

TABLE 5: COMPARISON OF ECONOMIC INDICATORS BETWEEN APPRAISAL AND COMPLETION

	EIRR	NPV		BCR	
Discount Rate	**NA**	**12%**	**6%**	**12%**	**6%**
At Appraisal: Original	75.1%	$414.3	NA	2.7	NA
Recalculated	25.9%	$239.7	NA	2.0	NA
At Completion					
Expected Value	53.7%	$394.8	$1,121.6	5.3	7.8
Standard deviation	0.04	55.9	147.6	0.6	0.9
Minimum	47.4%	$300.8	$863.9	4.3	6.2
Maximum	59.8%	$496.1	$1,377.9	6.4	9.3
Coefficient of Variance	0.06	0.14	0.13	0.12	0.1
Probability of low* outcome	0.0%	0.0%	0.0%	0.0%	0.0%

*low: < 12% IRR, < 0 NPV, <1 BCR

9. **At the completion of the project, the EIRR is estimated to be 53.7 percent.** The NPV is estimated to be between US$385 and US$1,122 million, and the BCR is estimated to be between 5.3 and 7.8, assuming 12 and 6 percent discount rates, respectively. Based on the recalculated efficiency matrix at appraisal, the efficiency matrix at completion shows some improvements resulting primarily from cost savings.

ANNEX 5. BORROWER, CO-FINANCIER AND OTHER PARTNER/STAKEHOLDER COMMENTS

1. This section contains ad verbum comments received by the IEs based on the final ICR draft.

BANGLADESH METEOROLOGICAL DEPARTMENT

2. We appreciate the comprehensive and well-structured ICR, which thoroughly addresses all facets of BWCSRP, emphasizing the "Lessons Learned" alongside the project's outcomes across physical, environmental, social, and financial domains.
3. Bangladesh Meteorological Department (BMD) extends its sincere thanks to the World Bank and collaborating partners for their instrumental role in the successful delivery of BWCSRP. This project has been considered as fundamental in advancing Bangladesh's meteorological and climate service infrastructure and thereby enhancing the country's ability to anticipate and respond to climate variability and extreme weather events.
4. BMD recognizes that through BWCSRP, significant enhancements were made in the country's weather observation networks, high-performance computing systems, forecasting capabilities, and integration of advanced weather forecasting models. Significant improvement of the ICT infrastructure has been ensured which has been crucial to enhance forecasting capabilities. These improvements have markedly increased the accuracy and lead time of weather and climate advisories, which are essential for protecting lives, livelihoods, and national development interests.
5. The project's emphasis on human resource development has notably strengthened the skills and expertise of BMD personnel across technical and operational domains. Furthermore, BWCSRP's regional cooperation framework has promoted robust information exchange and collaborative approaches with neighboring countries, fostering greater resilience against shared climate challenges.
6. BMD highly appreciates the World Bank technical team's expert input and sustained advisory support, which have been invaluable given the department's current limitations in specialized manpower and exposure to cutting-edge technologies—particularly in relation to the ICT advancements achieved through this project.
7. We appreciate the Task Team Leaders (TTLs) for their unwavering dedication and timely, constructive guidance that motivated the project team to uphold high standards in implementation and compliance with global best practices.
8. During the initial stages of the project, BMD encountered delays and challenges in the procurement process, which affected implementation timelines. However, with the World Bank's proactive support in expediting procurement-related queries and providing timely guidance, these hurdles were effectively addressed. This collaboration ensured smoother and more efficient procurement procedures, enabling BMD to successfully overcome those obstacles and allowed the project to be on track.
9. To maximize and sustain the long-term impact of BWCSRP, ongoing collaboration between BMD, BWDB, and key stakeholders will remain essential. Continued support from the World Bank in fostering stakeholder engagement platforms and regional knowledge networks will be vital.
10. […] It is important to note that several critical packages were deferred during the implementation stage due to time and manpower constraints; these should be prioritized and effectively executed during the forthcoming second phase. Additionally, continued investment in upgrading technical infrastructure, capacity building, and data

integration frameworks will enable BMD to sustain and expand its climate service delivery, thereby supporting Bangladesh's climate resilience agenda. I personally believe that the implementation of BWCSRP Phase-II is crucial to maintain this momentum and further advancing meteorological infrastructure and capabilities.

BANGLADESH WATER DEVELOPMENT BOARD

11. The Implementation Completion and Results (ICR) report is well-written, providing a clear summary of the project's achievements while also acknowledging areas for improvement in future similar initiatives.
12. BWCSRP has significantly enhanced the hydrological infrastructure with a noticeable capacity enhancement of IT-ICT infrastructure at the BWDB Data Centre, enabling BWDB to provide more accurate and timely forecasts. BWDB now has access to a Real-Time Data Acquisition System (RTDAS), which has profound implications for improving the quality of hydrological data and supports the newly upgraded nationwide flood model operated by the Flood Forecasting and Warning Center (FFWC). These advancements are vital to BWDB's operational responsibilities, particularly in flood control, irrigation planning, and the management of water resources, water infrastructure amid increasingly variable climate conditions.
13. The project provided extensive training on the operation and application of the advanced equipment procured under BWCSRP. End users were provided with standard operating systems (SoPs) with academic and hand on field trainings. Staff were also trained in utilizing the newly developed models, including those for salinity, drought, morphological changes, and ground water assessment, thereby enhancing their technical capacity. These trainings have been crucial in familiarizing staff with up-to-date technologies and ensuring effective use of modern tools.
14. We disagree with the "Satisfactory" rating given the project's accomplishments. We would argue that the rating can be improved to "Highly Satisfactory" as the project achieved all of its revised key PDO indicator targets.
15. BWDB sincerely appreciates the World Bank's technical team for their continuous guidance throughout the project implementation period. The Bank's role in steering the project through the identification, preparation, and supervision phases was highly effective, bolstered by constructive feedback from implementation support missions and timely assistance on critical matters. The World Bank team's field visits offered objective and valuable feedback that informed necessary remedial actions. As the project is of a high end technology, regular monitoring and assistance from provided from the Banks team and experts were very much effective and necessary without which it was not possible for such a successful achievements and completion of the project.
16. While implementation faced challenges such as data sharing constraints and coordination complexities among various agencies, BWDB recognizes the adaptive management approaches that addressed these issues effectively. Nonetheless, greater flexibility in procurement processes and more timely responses on administrative matters from the Bank could have further expedited project delivery.
17. To further enhance collaboration and project outcomes, BWDB suggests that the World Bank could improve by streamlining procurement procedures to reduce delays and by providing more timely responses on administrative and technical queries. Increased flexibility and faster decision-making would help accelerate implementation, ensuring smoother progress and greater efficiency in future projects.

18. Continued collaboration with the BMD and other stakeholders to ensure the sustained impact of BWCSRP's achievements is of paramount importance to us. The Bank's facilitation of stakeholder coordination platforms and promotion of regional knowledge exchange will further enhance collaborative efforts.
19. The establishment of the automated network and the development of new models have laid a strong foundation for hydrological advancement in our country. However, further efforts are required, including the additional installation of automated stations, continued refinement of models incorporating the latest data, and the establishment of a reliable early warning system. In this context, timely implementation of the second phase of BWCSRP is strongly encouraged to sustain and build upon these achievements. Last but not the least, Bank should keep an eye and be in touch with BWDB for the continued O&M of the newly established hydro-met systems.

DEPARTMENT OF AGRICULTURAL EXTENSION

20. […] The ICR report seems to offer a thorough and impartial summary of the Bangladesh Weather and Climate Services Regional Project. The document is factual and provides a clear overview of key milestones, accomplishments, outputs, challenges, and the approaches taken to address them.
21. BWCSRP has significantly advanced the capacity to generate and deliver tailored agrometeorological advisories. The enhancement of tools such as the Bangladesh Agro-Meteorological Information System (BAMIS) portal and the integration of community radio broadcasting have substantially improved the reach and effectiveness of climate services among farmers, contributing to more informed decision-making in crop management.
22. The project's focus on improving communication channels between meteorological data providers and end-users has been instrumental in bridging knowledge gaps. The use of community radio as a platform for localized weather forecasts and agricultural advisories has empowered rural farmers, particularly in remote areas, to better adapt to climate variability and reduce climate-induced risks.
23. DAE values the continuous technical guidance and expertise provided by the World Bank's technical team and the constructive feedback of the Task Team Leaders. Given DAE's operational challenges related to limited specialized manpower and the need for up-to-date technology, the support in modernizing ICT infrastructure and capacity development under BWCSRP has been particularly impactful.
24. We respectfully contest the "Satisfactory" rating assigned to the project's performance. The Project Development Objective has been successfully achieved, with weather and climate information systems performing reliably and the dissemination of agrometeorological advisories being conducted smoothly. Therefore, we believe the evaluation warrants an upgrade to "Highly Satisfactory."
25. DAE gratefully acknowledges the efforts of the Bank in initiating the process of data sharing with BMD and BWDB essential for effective agrometeorological services. We look forward to strengthening this cooperation further, fostering robust partnerships that will underpin sustainable, climate-smart, and resilient agricultural development in Bangladesh.
26. The sustainability of the project depends on a combination of financial, technological, institutional, environmental, social, and policy factors. By promoting inclusive, data-driven, and sustainable agricultural practices, the project can make a substantial contribution to climate resilience, agricultural productivity, and environmental conservation. The key to long-term success lies in continued investment, effective collaboration, capacity building, and a robust policy

framework. DAE strongly urges the World Bank to prioritize and expedite the implementation of the second phase of BWCSRP without any further delay to ensure the momentum gained is maintained and expanded.

APPENDIX 1. CLIENT SUMMARIES

1. This appendix provides shortened summaries from the Client that were provided as stand-alone documents or as part of Government-side reporting via Project Completion Reports (PCRs), which were prepared for each component separately. No joint Client ICR was submitted due to the similarity of government-prescribed PCRs.

COMPONENT A: STRENGTHENING METEOROLOGICAL INFORMATION SERVICES

2. **The BMD has significantly enhanced its weather forecasting capabilities between 2018 and 2024.** The resolution of the Weather Research and Forecasting model has improved from 18 to 3 km, enabling forecasts with a three-day lead time and two-day outlook compared to the previous 24-hour forecasts. Standard Operating Procedures (SOPs) and survey tools have been developed to streamline operations.
3. **BMD has upgraded its forecasting infrastructure by installing 125 Agrometeorological Automatic Weather Stations, 35 Synoptic AWS, and 65 Automatic Rain Gauges nationwide.** Advanced Weather Observation Systems were deployed at Dhaka, Chattogram, and Sylhet international airports. Implementing a high-performance computing system has also enabled more frequent model runs, improving forecast accuracy. A Decision Support System for drought and extreme events has also been introduced. Expanding observational networks and ICT infrastructure has enhanced BMD's ability to deliver accurate forecasts, supporting disaster preparedness and climate resilience. Installing the Climate Data Management System (CLIDATA) has facilitated efficient data acquisition, quality control, and analysis.
4. **Higher education degree programs were established to address capacity constraints and ensure a qualified workforce.** BMD has contributed to the establishment of the Department of Meteorology at the University of Dhaka, providing essential equipment and consultancy services to develop skilled professionals in meteorological services and offering new degree programs for meteorologists.
5. **Project implementation faced delays due to the COVID-19 pandemic, site selection difficulties, and administrative hurdles in securing land use agreements.** The challenges were addressed through virtual coordination, stakeholder engagement, and effective project management. Contract management issues were successfully addressed with guidance from the World Bank.
6. **Future project planning should incorporate time-bound work plans with contingency strategies.** Land acquisition and inter-agency agreements should be initiated early in the project cycle. Additionally, maintenance contracts should be clearly defined in procurement documents to ensure long-term sustainability, as tested and implemented under this Project.
7. **As this was BMD's first World Bank-financed project, some initially planned activities were postponed due to delays.** Following the project's successful completion, BMD recommends a follow-up phase to integrate deferred activities and meet emerging stakeholder demands.

COMPONENT B: AGRO-METEOROLOGICAL INFORMATION SYSTEMS DEVELOPMENT

8. **BWDB reported that the Project has significantly modernized Bangladesh's hydrological monitoring and flood forecasting capacities.** Before the Project, much of BWDB's network was outdated and manually operated, limiting the accuracy and timeliness of data. The Project transformed operations by installing over 1,500 automated

monitoring stations for groundwater, surface water, rainfall, and coastal data, along with ICT integration. Flood forecasting services were expanded from 31 to all 64 districts, and a 2D dynamic flood inundation modeling capability was introduced for the entire country, including coupled models for coastal flooding. The Project also strengthened BWDB's GIS and survey capacities, resulting in the development of a highly detailed national Digital Elevation Model (2 x 2m resolution) and improved hydrological and morphological survey capabilities through new equipment and training.

9. **BWDB emphasized that the Project enabled the diversification of forecast products, including new models for groundwater, salinity intrusion, sediment transport, riverbank erosion, and drought prediction.** Critical ICT systems were upgraded by installing modern servers, network equipment, and software for data acquisition, management, and dissemination, including mobile applications and web portals for public information access. Community-based early warning systems were introduced in two flood-prone districts, and significant advances were made in integrating hydrological data with forecasting services to enhance disaster preparedness across the country.
10. **Implementation faced key challenges, including COVID-19 disruptions, complex site selection for remote coastal monitoring stations, and negotiating MoUs with multiple agencies for infrastructure use.** BWDB noted that proactive coordination, technical flexibility, and virtual supervision enabled the Project to overcome these barriers. Post-project sustainability challenges were identified, particularly regarding operation and maintenance funding, mobile network stability, and protection against theft and natural hazards. However, BWDB secured initial maintenance budgets and contracts and developed an experimental electric sentry system to enhance station security.
11. **BWDB emphasized key lessons learned, including the importance of early technical training for project teams and promptly approving institutional agreements for project momentum.** The agency further highlighted the critical need for continuous maintenance and technical upgrades to ensure the sustainability of high-end hydrological systems.

COMPONENT C: AGRO-METEOROLOGICAL INFORMATION SYSTEMS

12. **Agromet services in South Asia have a longstanding history, with India playing a pivotal role in providing weather forecasts for agriculture.** The India Meteorological Department (IMD), established in 1875, was founded in response to the country's experiences with tropical cyclones and famines. Pakistan followed suit by creating the Pakistan Meteorological Department (PMD) after the partition of India, while the Bangladesh Meteorological Department (BMD) was formed post-liberation in 1971. The BMD's primary responsibility is monitoring weather patterns, issuing forecasts, and providing early warnings for extreme events.
13. **India has highly developed meteorological services, while Bangladesh faces significant constraints.** The BMD has a dedicated agromet division that provides forecasts for agricultural planning, including long-range and seven-day weather outlooks. Since 2017, the DAE in Bangladesh, with World Bank funding, has been involved in the Agro-Meteorological Information Systems Development Project (AMISDP)[12] to enhance climate services for agriculture. Integrating sub-seasonal and seasonal forecasts into agriculture is still in its early stages.

[12] Client-side name for Component C.

14. **Bangladesh's agricultural sector is susceptible to weather variations, with extreme events causing substantial losses.** For example, the 2007 cyclone Sidr caused damage amounting to US$1.7 billion, representing 2.6 percent of the country's Gross Domestic Product. Climate projections indicate a 20–30 percent increase in temperature and rainfall by 2100. Despite these challenges, Bangladesh's hydromet infrastructure remains weak. In response, the Government of Bangladesh, through the Ministry of Agriculture and the DAE, launched the AMISDP in 2016, which aims to improve climate resilience in farming communities. The key objectives were the following:
    (a) *Enhancing Agro-Meteorological Services:* Providing timely information to farmers to help them manage extreme weather events and improve agricultural productivity.
    (b) *Developing a Science-Based Information System:* Creating an agrometeorological system to provide accurate data on crop yields, land holdings, and weather forecasts.
    (c) *Disseminating Agro-Meteorological Information:* Establishing communication channels, including digital platforms, to deliver vital information to farmers.
    (d) *Building Capacity at All Levels*: Strengthening national and local authorities to deliver climate information services effectively.
    (e) *Enhancing Climate Resilience:* Establishing a sustainable early warning and risk management system for agriculture.
15. **Bangladesh's susceptibility to weather extremes has highlighted the need for robust hydro-meteorological infrastructure.** In response, under the Ministry of Agriculture, the DAE launched the AMISDP as part of the Bangladesh Weather and Climate Services Regional Project (BWCSRP), funded by the World Bank. The Project's goal was to establish a science-based, digitized agromet database, including data on land holdings, cropping systems, and average crop yields across 495 Upazilas.
16. **The AMISDP has made significant strides in achieving its objectives.** The BAMIS portal, which has been operational since 2019, facilitates collecting and distributing agromet data to farmers. Dissemination channels include digital display boards, SMS to 30,000 lead farmers, and community radios. BWDB is expanding its flood forecasting network, and the BMD has installed 125 ARGs and expanded its forecast products. The DAE has assisted farmers with crucial crop management decisions and employs various communication methods, including BAMIS, mobile apps, community radios, digital display boards, and SMS/IVR voice messages.
17. **Key achievements of the AMISDP include:**
    (a) *National and District-Level Agromet Advisories:* DAE provides weekly agromet advisories at the national level and bi-weekly at the district level.
    (b) *Infrastructure Development:* The project established 65 digital display boards, 488 agromet kiosks, 13 community radio stations, and a robust BAMIS portal.
    (c) *Data and Modeling Support:* The Project developed crop simulation models, digitized historical and current data, and created agromet databases for 495 Upazilas.

(d) *Capacity Building:* The Project supported the creation of agromet departments at the Bangladesh Agricultural University (BAU) and Bangabandhu Sheikh Mujibur Rahman Agricultural University (BSMRAU) and trained stakeholders.

18. **The BWCSRP activities are progressing well after the closing of the Project[13], with positive feedback from users at the grassroots level.** However, continuing and expanding activities in a potential second phase would be crucial for strengthening agromet services in rural Bangladesh. The second phase of the AMISDP is expected to generate significant benefits beyond agriculture, contributing to environmental sustainability, social equity, and economic development, further enhancing overall societal advancement.

[13] The summary was submitted to the World Bank in March 2025.

APPENDIX 2. EXTENDED LIST OF LESSONS LEARNED

1. **Despite their importance, national meteorological and hydrological services (NMHSs) are underdeveloped or underutilized in many countries, facing significant scaling challenges.** Weather, water, and climate affect almost all aspects of life and sectors of the economy to some extent. NMHSs are in charge of providing relevant information to enable timely action. However, globally, many NMHSs are poorly equipped to fulfill this key task. The World Bank has supported countries in enhancing hydromet service provision through modernization processes. However, such processes are very challenging due to:
   (a) *Adoption Challenges.* Rapidly evolving technology requires institutional changes and adaptation.
   (b) *System Integration.* Merging old and new systems from various sources to ensure reliable outputs.
   (c) *Environmental and Security Risks.* Outdoor installations face theft, vandalism, disasters, and land disputes.
   (d) *Resource Constraints.* Limited spare parts, technical support, high maintenance costs, and staff turnover.
   (e) *Communication and awareness gaps arise from unreliable networks, limited ICT capacity, and outdated public service dissemination.*
2. **The task team of BWCSRP is producing a technical note on lessons learned, which is summarized in this section to ensure that relevant technical recommendations are accessible to other teams.** This note provides insights into successes and learning moments resulting from BWCSRP and presents key lessons learned to inform a potential next phase of the Project and other hydromet projects. A sub-set is covered in the lessons learned section of the main text, while a more extensive technical edition for publication is under preparation.

PROJECT DESIGN

3. **Expanding Hydromet Services to New Sectors and Geographic Areas.**
   (a) *Context:* While the Project significantly strengthened the core hydromet infrastructure, additional improvements are required to extend forecast lead times, enhance coverage in *coastal and vulnerable areas*, and develop tailored services for new sectors, including *health, transport, and urban planning*. The demand for expanded hydromet services is reflected in government requests for a *second phase of investments*, highlighting continued interest from key agencies.
   (b) *Recommendation:* Hydromet modernization efforts should focus on expanding coverage to underserved areas, particularly coastal regions prone to extreme weather events. Tailored forecasting services should be developed for high-priority sectors such as health (heat stress and mosquito-borne diseases), transport, and urban planning. Future project phases should align with national priorities and sectoral development strategies to maximize impact.
4. **Ensuring Adequate Implementation Time for Hydromet Modernization.**
   (a) *Context:* The Project was extended twice – first for 12 months and then for another 11 months – to compensate for delays in effectiveness and COVID-19 disruptions. A longer implementation timeline is crucial due to the complexity of hydromet modernization. This ensures proper integration of system components and adequate

supervision of initial operations. Experience from regional and global projects indicates that an *eight-year timeframe is the minimum required for the first phase of hydromet system modernization.*

(b) *Recommendation:* Future hydromet modernization projects should adopt a realistic implementation timeline of at least eight years for the initial phase. Project designs should incorporate flexibility for *contingencies such as procurement delays, system integration challenges, and unforeseen external shocks.* Extension mechanisms should be clearly defined to allow for necessary adjustments without compromising project outcomes. Another option is to design a multiphase programmatic approach, building capacity and systems sequentially.

PROJECT IMPLEMENTATION

5. **Institutionalizing Inter-Ministerial Coordination for Sustainability.**

(a) *Context:* An *inter-ministerial meeting*, chaired by the *External Relations Department*, was held during the Project's closing mission to assess achievements and ensure sustainability. Key commitments included securing *operational and maintenance (O&M) budgets*, finalizing *pre-negotiated annual maintenance contracts*, and *retaining trained staff* in key positions. These steps were recognized as essential for long-term sustainability.

(b) *Recommendation:* Future projects should institutionalize *high-level inter-ministerial coordination mechanisms* to ensure the sustainability of hydromet investments. Annual or biannual *ministerial-level review meetings* should be convened to assess progress, address challenges, and reinforce institutional commitments. Sustainability measures – including *O&M budget allocation, long-term maintenance contracts, and staff retention strategies* – should be embedded in project planning and implementation frameworks.

6. **Active and Continuous Training.**

(a) *Context:* The Project included extensive training programs, with 42 seminars and conferences conducted. However, high staff turnover and insufficient retention strategies have led to gaps in institutional knowledge. This situation necessitates repeated capacity-building efforts.

(b) *Recommendation:* Future projects should implement structured, ongoing training programs with built-in mechanisms for knowledge transfer. Training-of-trainers models should be adopted to ensure continuity. Staff turnover should be managed strategically, limiting transfers to no more than 20% annually to retain institutional memory. Additionally, training programs should be role-specific, ensuring participants gain skills directly relevant to their responsibilities.

7. **Surveys for User Feedback.**

(a) *Context:* The Project developed user satisfaction survey tools to assess service effectiveness. However, these tools were only applied at the end of the Project, limiting their utility in making real-time adjustments.

(b) *Recommendation:* Future projects should integrate continuous feedback mechanisms, conducting periodic surveys throughout implementation. User insights should drive continuous improvement in service delivery. This ensures that hydromet products remain relevant and user-friendly. Feedback mechanisms should be embedded within digital platforms, allowing for real-time monitoring of user experiences.

8. **System Integration.**

(a) *Context:* One of the significant challenges in the Project was integrating meteorological, hydrological, and agricultural data across multiple agencies. Data-sharing agreements took years to finalize, delaying full interoperability between systems.

(b) *Recommendation:* Future projects should establish inter-agency data-sharing frameworks early in the project cycle. Standardizing data formats and protocols is essential for achieving seamless integration. Clear institutional mandates should define data-sharing responsibilities, and technical coordination between vendors and implementing agencies should be strengthened to prevent compatibility issues.

9. **Addressing Procurement Challenges Affecting Project Sustainability.**

(a) *Context:* Some institutions could not sign contracts extending beyond the project period, leading to maintenance gaps. Additionally, budgetary limitations prevented long-term commitments, such as funding SIM cards required for data transmission.

(b) *Recommendation:* Future projects should be designed with flexibility to accommodate long-term operational needs. Approval processes should be streamlined, and project structures should include clear financial planning for post-project sustainability.

10. **Dissemination, Both High-Level and Local, is Critical.**

(a) *Context:* Despite improvements in hydromet services, public awareness and accessibility remained limited. DAE used several successful citizen outreach channels. However, while BWDB successfully launched a flood forecasting app and ran a television campaign, user adoption was low due to inadequate outreach. Similarly, BMD's weather forecasts remained outdated in format and lacked engagement strategies to attract younger audiences.

(b) *Recommendation:* Future projects should prioritize structured communication strategies. Public awareness campaigns must leverage social media, mobile apps, and traditional media to maximize outreach. Hydromet agencies should modernize weather forecasts with user-friendly formats and interactive features to enhance engagement. Feedback loops should be established to refine dissemination strategies based on user needs.

11. **Purchase of HPC with a Performance-Based Approach.**

(a) *Context:* The Project successfully implemented a performance-based procurement model for High-Performance Computing (HPC) systems. Instead of solely relying on technical specifications, bid evaluations included computational speed and energy efficiency criteria, ensuring cost-effectiveness while maximizing long-term operational efficiency.

(b) *Recommendation:* Future projects should replicate this performance-based procurement model for complex ICT infrastructure. Procurement contracts should include measurable performance indicators, such as computation time and power consumption, incentivizing vendors to provide high-efficiency systems (Table 6). Pre-bid meetings and sample performance tests should be incorporated to ensure transparency and vendor accountability.

TABLE 6: EXAMPLE OF PERFORMANCE CONDITIONS FOR HPC PURCHASE

Simulation Time	Lowering in bid evaluation price
8+ minutes	Bid rejected
<8 minutes	0 percent (no incentive but bid valid)
<7 minutes	5 percent
<=6 minutes	10 percent (no additional incentive beyond the minimum time specified)

12. **Developing Standard Operating Procedures (SOPs) for Hydromet Modernization.**
   (a) *Context:* The Project highlighted the complexity of integrating multiple hydromet system components, requiring clear *Standard Operating Procedures (SOPs)* to guide future modernization efforts. These SOPs should define *technology adoption protocols, system integration workflows, and long-term sustainability measures.*
   (b) *Recommendation:* Future hydromet modernization projects should develop *comprehensive SOPs* to clarify roles, responsibilities, and operational processes across implementing agencies. A *National Hydromet Roadmap* should be developed, outlining targets for modernization over a *5–10-year period*. SOPs should also guide *system integration, technology procurement, and human resource development* to strengthen long-term operational capacity.
13. **Strengthening Site Selection Processes for Hydromet Installations.**
   (a) *Context:* The Project faced challenges due to inadequate site selection, leading to issues related to land availability, instrument security, mobile network coverage, and meteorological feasibility. Poorly chosen sites resulted in delayed installations, theft, and suboptimal data collection, emphasizing the need for a more structured approach.
   (b) *Recommendation:* Future projects should establish a *systematic site selection process* before awarding contracts. This process should include *field assessments, community consultations, and detailed site selection reports* covering geographic suitability, land tenure security, and network connectivity. Adopting a *structured multi-criteria site selection framework* will minimize risks and optimize system performance.
14. **Enhancing Security Measures to Prevent Theft and Vandalism.**
   (a) *Context:* Theft and vandalism of meteorological equipment posed significant risks to project success. Sites located on *government premises or within staffed facilities* were found to be more secure, while installations in *remote or unmanned locations* faced higher risks.
   (b) *Recommendation:* Future projects should prioritize *installations within secure government facilities* or staffed locations to minimize theft and vandalism risks. Establishing *data-sharing agreements* where installations occupy other agencies' premises will enhance local ownership. Security measures should be enhanced through *community engagement, police collaboration, and public awareness campaigns* highlighting the benefits of hydromet systems in disaster prevention and climate resilience.
15. **Ensuring Availability of Spare Parts and Technical Support.**
   (a) *Context:* The Project underscored the need for a *long-term strategy* for spare parts and technical support to prevent disruptions. Equipment failures without timely replacements can *compromise data continuity* and reduce the effectiveness of hydromet systems.

(b) *Recommendation:* Future projects should establish a comprehensive spare parts management strategy, ensuring in-house maintenance capacity or long-term agreements with manufacturers. Procurement contracts should include predefined pricing mechanisms for spare parts and impose penalties for repair delays. A digital asset management system should be implemented to track equipment health and vendor performance in real-time.